nature communications

Article

https://doi.org/10.1038/s41467-026-73721-w

Pan-cancer single-cell atlases of mouse and human tumor-associated dendritic cells

Received: 12 November 2025

Accepted: 19 May 2026

Published online: 30 May 2026

Check for updates

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Dendritic cells (DCs) are critical inducers of anti-tumor immunity. To achieve a comprehensive mapping of mouse and human DC subsets and states in a cancer context, here we generate pan-cancer mouse and human tumor-associated DC (TADC) scRNA-seq atlases, encompassing 14 mouse tumor models and 10 human cancer types, within which we identify several lineage-defined DC subsets along with maturation/functional states. We show that TADCs acquire an inflammatory profile with tumor progression and that tumor-mediated reprogramming occurs within the DCs from lymph nodes of tumor-bearing mice. Importantly, we demonstrate that TADCs are broadly conserved between mice and humans, although species-specific differences may exist in some subsets and states. Moreover, we present a comprehensive assessment of how different human TADC clusters associate with patient survival outcomes. Overall, we provide an in-depth characterization of the TADC compartment in mouse and human cancers, which can improve our understanding of the tumor microenvironment and contribute to the development of new anti-cancer therapies.

Dendritic cells (DCs) are the most efficient antigen-presenting cells (APCs), driving T-cell responses[1]. Based on their ontogeny, DCs can be subdivided into two main lineages: the conventional or classical DCs (cDCs, also called myeloid DCs), and the plasmacytoid DCs (pDCs). Although currently debated, several fate-mapping studies have recently added DC3 as a third potential lineage to the DC family[2]. Hence, the prevailing view in the field is that DCs can be subdivided into three lineages: cDCs, pDCs, and DC3[3,4]. In the bone marrow (BM),

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cDC and monocytes share a common progenitor, called the monocyte-DC progenitor (MDP)[5–7]. While MDP-derived common DC progenitors (CDPs) give rise to cDC[8–10], MDP can differentiate into monocytes[5,11], although monocytes are also known to arise from the granulocyte-monocyte progenitor (GMP)[12,13]. The cDC lineage can be classified into Clec9a$^+$Xcr1$^+$ cDC Type 1 (cDC1) and CD11b$^+$Sirpα$^+$ cDC Type 2 (cDC2)[14]. In the BM and the periphery, several DC progenitors, each leading to specific DC lineages, have been discovered to date, thereby highlighting the intricate diversity of fates of the mature peripheral DC subsets. A part of the cDC2s are derived from CDPs via their immediate precursors, the pre-cDC2s[15]. Similarly, cDC1s differentiate from pre-cDC1s[15]. Interestingly, a cell population with progenitor potential exhibiting shared features of pDCs and cDC2s that can contribute to a part of the pDC and cDC2 compartments has been described[3,16–19]. These cells have been addressed by a plethora of names, including CX3CR1$^+$CD8a$^+$ DC[20], noncanonical DC[21], AXL$^+$ DC[22], pre-DC[23], AXL$^+$SIGLEC6$^+$ (AS) DC[24], pDC-like cells[3,25], and transitional (tDC)[16,17], but it is unclear to what degree they overlap.

Several studies have described multiple cDC2 subsets/states in homeostasis and inflammation. cDC2s were initially classified into Notch2-dependent Esam^{hi} and Klf4-dependent cDC2s[26,27]. Subsequently, based on Tbet-expression, cDC2s were further categorized into CLEC12A$^-$Esam$^+$T-bet$^+$ cDC2As and CLEC12A$^+$Esam$^-$T-bet$^-$ cDC2Bs[28]. Recently, pre-cDC2As[29] and pre-cDC2Bs[25,29] have been described downstream of pre-cDC2s to give rise to cDC2As and cDC2Bs, respectively[29]. The DC3s have been described in mouse and human blood, BM, and spleen as a lineage that is neither CDP- nor cMoP-derived but rather arise from an intermediate Ly6C$^+$ MDP progenitors through pro-DC3s[2,30,31].

In addition to being defined by their lineage, DC subsets can attain different maturation/functional states during their journey in the periphery. For instance, upon stimulation, both cDC1s and cDC2s can achieve a mature state referred to as the CCR7$^+$ DCs[32–34] (also known as migratory DC (migDC)[32,35], matureDC[36], late mature DCs[37], mregDC[33], DC3[38,39], and LAMP3$^+$ DC[40,41]).

Both cDC1s and cDC2s are critical in inducing anti-tumor immunity[42–46]. Multiple studies have thoroughly characterized the cDC1 population at both the phenotypic and functional levels[47], highlighting that they represent a rather uniform cell compartment. cDC1s are proficient in cross-presenting exogenous antigens to CD8$^+$ T cells and are critical in the rejection of highly immunogenic murine tumors[48]. Moreover, cDC1s are potent regulators of cancer immunotherapy (anti-4-1BBL, anti-PD-1, anti-CTLA-4, and adoptive T-cell therapy) responses, as demonstrated in transplantable mouse tumor models[49–51]. Alternatively, cDC2s show a greater phenotypic diversity in different inflammatory settings, including cancer, and can induce Th1, Th2, Th17, Tfh (T follicular helper), and Treg responses[42,43,46,47]. Additionally, a state of cDC2s, namely IRF8$^+$CD64$^+$ inflammatory cDC2s (inf-cDC2s), characterized by the expression of interferon-stimulated genes (ISGs), was described to orchestrate CD4$^+$ and CD8$^+$ T-cell immunity in respiratory viral infections[52]. In cancer, cDC2s expressing high levels of ISGs can prime CD8$^+$ T cells by acquiring and presenting tumor-derived antigen-MHC-I complexes, a process known as MHC cross-dressing[42]. More recently, cDC2 were also shown to be able to cross-prime CD8$^+$ T cells through cross-presentation of antigens in the form of antigen-antibody complexes[53]. Upon treatment of tumor-bearing mice with anti-CD40, cDC2s potently activated CD8$^+$ T cells in the absence of cDC1[43], demonstrating the broad range of functions cDC2 can perform.

A major challenge in elucidating the diverse functions of cDC2s and harnessing their therapeutic potential lies in the fact that cDC2s encompass multiple subsets and functional states. To date, the majority of cDC2/DC3 subsets/states identified have been characterized under homeostasis or non-cancer inflammatory settings, limiting our understanding of their roles within a tumor context. Moreover, the few existing pan-cancer transcriptomic DC atlases are limited to human tumors and do not capture the full diversity of tumor-associated DC (TADC) subsets and states[32,38,54–57], nor do they allow direct cross-species comparisons. To fill this knowledge gap, a comprehensive analysis of the cDC2/DC3 landscape across both human cancers and preclinical mouse models is essential. Gaining deeper insight into the DC compartment across species is crucial for fully unlocking the therapeutic potential of DCs and opening new doors for the generation of anti-cancer DC-based therapies.

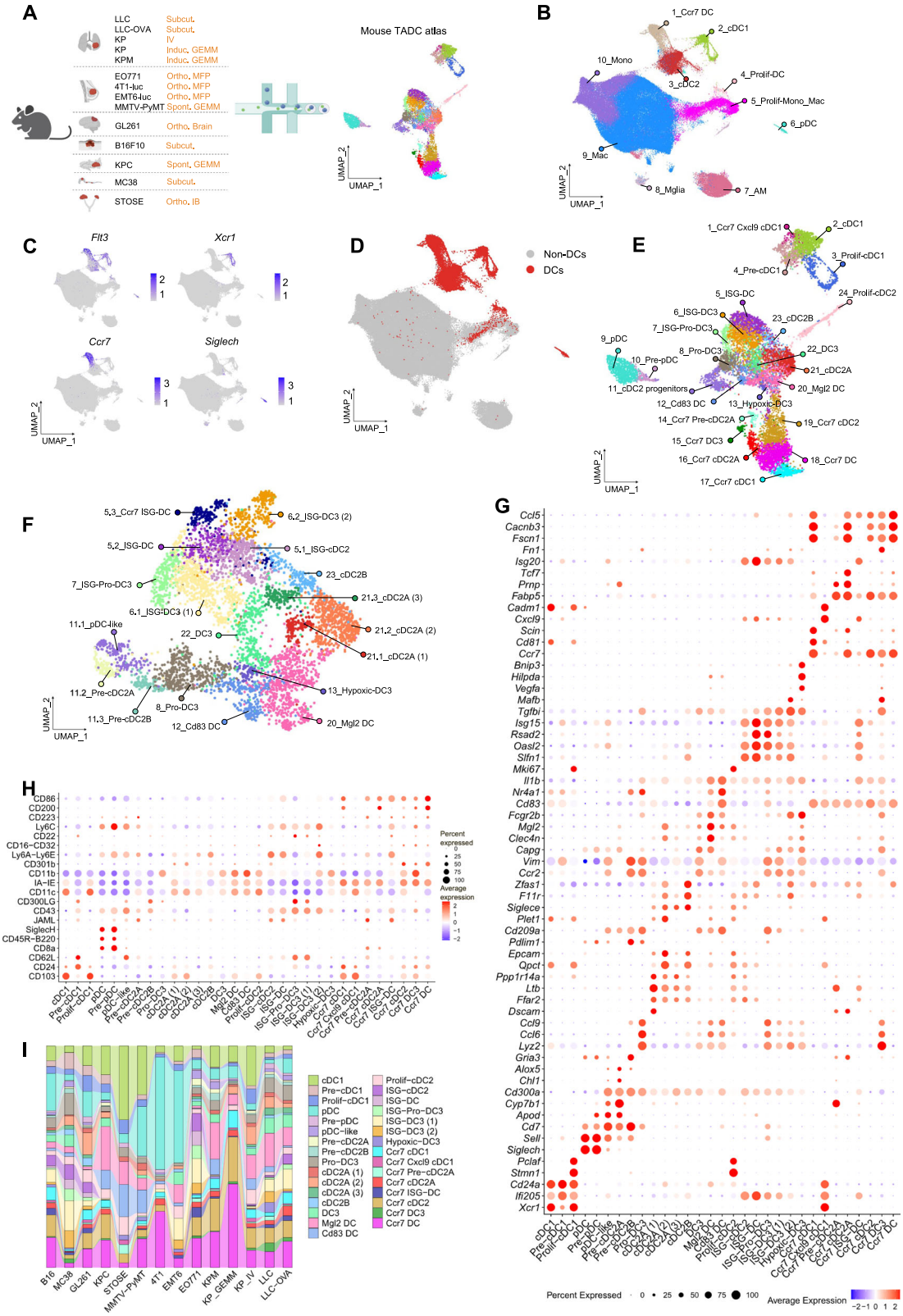
In this study, we generate pan-cancer mouse and human TADC single-cell RNA-sequencing (scRNA-seq) atlases to unravel the TADC heterogeneity across mouse and human, defining distinct DC lineages and maturation/functional states. We show that the distinct DC subsets/states evolve with tumor progression and demonstrate that the presence of a tumor induces inflammatory reprogramming in DCs in lymph nodes (LNs). Aiming to assess the mouse-to-human translatability of our data, we highlight the gene and predicted regulon activity correlation between mouse and human TADC subsets across different cancers. Finally, we assess correlations of our TADC subsets/states with patient outcome. Overall, we provide mouse and human TADC atlases, which can serve as valuable tools for characterizing the pan-cancer DC landscape across tumors, tumor-associated, and naive lymphoid organs. Our atlases are available for visualization, download, and further analysis from our interactive web server (www.single-cell.be/Laouimmunology/TumorDCAtlas).

Results

Single-cell RNA sequencing reveals conserved mononuclear phagocyte subsets and states in fourteen mouse cancer models

To generate a comprehensive transcriptional pan-cancer atlas of mouse TADCs, we aligned scRNA-seq and/or CITE-seq datasets from 14 preclinical mouse cancer models (Fig. 1A). The lung cancer models within the atlas included subcutaneous lung cancer (Lewis lung carcinoma (LLC)[43,58] and the more antigenic LLC-OVA models), inducible genetically engineered mouse model (GEMM) *Kras*$^{LSL\text{-}G12D/+}$;*Trp53*$^{fl/fl}$ (KP), inducible GEMM *Kras*$^{LSL\text{-}G12D/+}$;*Trp53*$^{fl/fl}$;*Msh2*$^{fl/fl}$ (KPM), and intravenously injected (IV) KP tumors[33]. Breast cancer models comprised EO771[59], 4T1-luc[60], and EMT6-luc[60] injected orthotopically in the mammary fat pad (MFP), as well as the spontaneous GEMM MMTV-PyMT[61,62]. Additionally, the atlas includes orthotopic glioblastoma (GL261)[35], subcutaneous melanoma (B16F10)[43], spontaneous pancreatic ductal adenocarcinoma (*Kras*$^{+/LSL\text{-}G12D}$;*Trp53*$^{+/LSL\text{-}R172H}$;*Pdx1*^{Cre} (KPC))[63], subcutaneous colon carcinoma (MC38)[64,65], and orthotopically injected STOSE[66] ovarian cancer model (Fig. 1A, Supplementary Data 1). First, we integrated 117,517 mononuclear phagocytes (MNP), including monocytes (Mono), macrophages (Mac), DCs, and pDCs from the different datasets using the Harmony package[67] (Fig. 1B, Supplementary Fig. 1A–C, 2A, B). The optimal integration strategy was determined based on the iLISI and cLISI metrics[67], measuring the integration efficiency and accuracy, as well as by qualitative assessment of UMAP diagnostic plots, aiming to achieve the best balance between increasing alignment strength and retaining cell type separation (Supplementary Fig. 1A–C). The MNPs were broadly subclustered into 10 clusters and annotated as DCs (Cl 1, 2, 3, 4 and 6), proliferating monocytes and macrophages (Prolif-Mono_Mac, Cl 5), alveolar macrophages (AM, Cl 7), microglia (Mglia, Cl 8), macrophages (Cl 9), and monocytes (Cl 10) (Fig. 1B). Among the monocytes, fine-grained subclusters were identified based on differentially expressed genes (DEGs) and surface proteins (based on CITE-seq), and included non-classical monocytes (Cl 10.2: NonClass-Mono - genes: *Treml4, Adgre4, Gngt2, Ace, Ear2;* proteins: CD43, Ly6C^{lo}), classical monocytes (Cl 10.3: Class-Mono - genes: *Chil3, Hp, Mgst1;* proteins: Ly6C^{hi}), and a monocyte cluster highly expressing interferon-stimulated genes (ISG) including *Isg15, Ifit204, Ifi209*, and *Ifit3* (Cl 10.1: ISG-Mono) (Supplementary Fig. 2A, Supplementary Fig. 2C, D, Supplementary Data 2–3).

https://doi.org/10.1038/s41467-026-73721-w



Additionally, 14 macrophage subclusters were identified, including Cxcl9 Mac (Cl 9.1: *Cxcl9, Gbp2, Slamf8*), ISG-Mac (Cl 9.2: *Ctss, Bst2, Isg15, Ifit3*), Clec4b1 Mac (Cl 9.3: *Clec4b1, Il10, Cd300e, Cd72*), Cx3cr1 Mac (Cl 9.4: *Cx3cr1, Pmepa1*), Folr2 Mac (Cl 9.5: *Folr2, Cbr2, Stab1, Selenop*), Ctsk Mac (Cl 9.6: *Ctsk, Tnfrsf11a, Bdh2, Slc9b2, Atp6v0d2*), large peritoneal macrophages (Cl 9.7: LPM; *Gata6, Icam2, Itga6, Timd4*), Hypoxic-Mac (Cl 9.8: *Vegfa, Bnip3, Hilpda, Hmox1)*, Spp1 Mac (Cl 9.9: *Spp1, Ccl9, Ccl6*), Mgl2 Mac (Cl 9.10: *Mgl2, Dcstamp*), Immature-Mac (Cl 9.11: *Ccr2, Ly6c2*^{lo}*, C1qa*^{lo}*, C1qb*^{lo}*)*, alveolar macrophage (Cl 7.1: AM; *Marco, Siglecf, Mrc1, Plet1*), Retnla AM (Cl 7.2: *Retnla, Timp1*) and microglia (Cl 8: Mglia: *Tmem119, P2ry12, Sparc, Sall1*) (Supplementary Fig. 2A, Supplementary Fig. 2C, D, Supplementary Data 2–3). Interestingly, Hypoxic-Mac showed high expression of lipid-associated macrophage (LAM) genes[68], including *Gpnmb, Fabp5*, and *Lgals3* (Supplementary Fig. 2E).

Fig. 1 | Single-cell RNA sequencing reveals 31 conserved TADC subsets and states in 14 mouse cancer models. A Schematic overview of the scRNA-seq datasets used to generate the mouse MNP and TADC atlases. Created in BioRender. Laoui, D. (2026) https://BioRender.com/zbzft9j. LLC $n = 2$, LLC-OVA $n = 1$, KP IV $n = 1$, KP Induc. GEMM $n = 1$, KPM $n = 1$, EO771 $n = 2$, 4T1-luc $n = 1$, EMT6-luc $n = 1$, MMTV-PyMT $n = 6$, GL261 $n = 4$, B16F10 $n = 2$, KPC $n = 4$, MC38 $n = 7$, STOSE $n = 1$. For details on number of mice pooled per sample refer to Supplementary Data 1. **B** UMAP plot of 117,517 tumor MNP atlas cells, colored by their major cell type. **C** UMAP plots highlighting DC-specific gene expression within the MNP atlas. **D** UMAP plot highlighting all DCs within the MNP atlas. **E** UMAP plot of 11,071 TADCs, clustered into major DC subsets and states. **F** UMAP plot of 5,055 cDC2/DC3 cells (including 21_cDC2A, 23_cDC2B, 22_DC3, 5_ISG-DC, 6_ISG-DC3, 7_ISG-Pro-DC3, 8_Pro-DC3, 11_cDC2 progenitors, 12_Cd83 DC, 13_Hypoxic-DC3, and 20_Mgl2 DC) subsetted and re-clustered from (**E**). **G** Dotplot showing the average expression of genes specific for the different TADC atlas clusters. **H** Dotplot showing the average expression of differentially expressed (DE) surface proteins, retrieved by CITE-seq of antibody-derived tags (ADT) within the different TADC atlas clusters. Since CITE-seq was not performed on all the datasets included in the mouse atlas, DE surface protein list was generated only using the CITE-seq datasets, which included MC38, GL261, EO771, KP (inducible GEMM and IV), and LLC datasets. **I** Alluvial plot showing the relative frequency of the TADC subsets and states (from **E** and **F**) across the different mouse cancer models. Subcut subcutaneously injected, Induc. GEMM inducible genetically engineered mouse model, Otho orthotopically injected, MFP mammary fat pad, Spont spontaneous tumor model, IB intrabursally injected, MNP mononuclear phagocyte, Mono monocyte, Mac macrophage, AM alveolar macrophages, Mglia microglia, ISG interferon-stimulated genes, Prolif proliferating.

Additionally, a cluster of Prolif-Mono_Mac (Cl 5: *Mki67*, *Stmn1*) was observed (Supplementary Fig. 2A, Supplementary Fig. 2C, Supplementary Data 2).

The different MNP clusters varied in abundance across cancer types (Supplementary Fig. 2F, G). NonClass-Mono were the most abundant in LLC, EO771, MC38, and B16 tumors compared to other cancer types, while a higher proportion of Class-Mono was present within LLC, LLC-OVA, and KPC tumors (Supplementary Fig. 2F, G). Cx3cr1 Mac were abundant in GL261 and MMTV-PyMT tumors, while ISG-Mono and ISG-Mac were most abundant in EO771 and MC38 tumors. Folr2 Mac, Spp1 Mac, and Hypoxic-Mac were most abundant in LLC, LLC-OVA, and STOSE tumors. Mgl2 Mac were predominantly found in KPC tumors. Evidently, tissue-resident macrophages were found in tumor types present in specific anatomic sites; LPMs were most abundant in KPC tumors, while Mglia were exclusively found in the glioblastoma model GL261. Similarly, AM were most abundant in orthotopic lung cancer models, including KP (GEMM, IV) and KPM (GEMM) models, while Retnla AM were restricted only to the GEMM models KP and KPM, and were absent in KP IV tumors (Supplementary Fig. 2F, G). Interestingly, Ctsk Mac were highly enriched in LLC tumors, but were also found in EMT6, STOSE and KPC tumors and resembled osteoclast-like cells, expressing *Ctsk* and *Tnfrsf11a* (encoding TNFRSF11/RANK), involved in facilitating bone metastasis of breast cancer via bone degradation[69–71], as well as *Atp6v0d2*, coding for a known regulator of osteoclast differentiation[72] (Supplementary fig. 2C, Supplementary Fig. 2F, G). Tumor-associated macrophages in human breast cancer have also been observed to differentiate into osteoclast-like cells[73]. Importantly, while we observed apparent differences in the abundance of MNP clusters across different mouse cancer models, these should be viewed as indicative trends, as limited data points in some cancer types precluded robust statistical analysis. Collectively, we have generated an MNP scRNA-seq atlas from 14 mouse cancer models showing the presence of multiple monocyte, macrophage, and DC subsets and states that vary in abundance across tumor types.

The mouse TADC atlas harbors ontogenically defined subsets and functional states

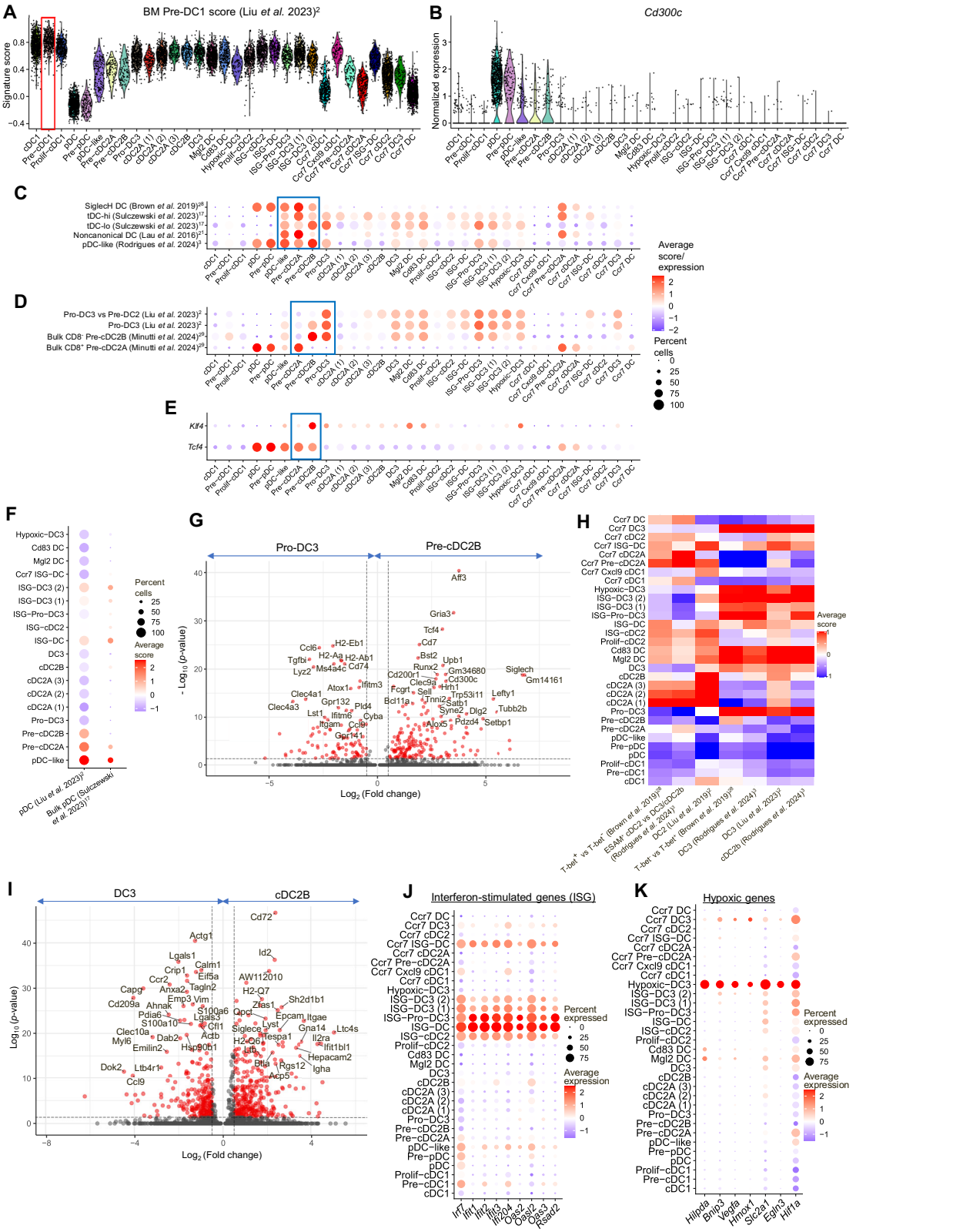
DCs within the MNP atlas were identified using general DC-specific surface proteins (including CD26, IA-IE, CD11c) and genes (including *Flt3*, *H2-Aa, H2-Eb1, H2-Ab1* (encoding MHC-II)) (Fig. 1C, Supplementary Fig. 2C, D, Supplementary Data 2–3). Based on known marker genes and surface proteins, the main DC subsets and states were identified, including cDC1 (genes: *Xcr1, Clec9a, Cd24a, Irf8;* proteins: XCR1, CD24, CD103), cDC2 (genes: *Cd209a, Klrd1, Clec10a, Ltb;* protein: CD301a), pDCs (genes: *Siglech, Ccr9, Ly6d*; proteins: CD45R-B220, SiglecH), and Ccr7 DCs (genes: *Ccr7, Fscn1;* proteins: CD200, CD274, CD80, CD86) (Fig. 1B, C, Supplementary Fig. 2C, D, Supplementary Data 2–3). To fine-map the DC heterogeneity, we subsetted and re-clustered cDC1s, cDC2s, Ccr7 DCs, pDCs, and proliferating DCs from the MNP atlas (Fig. 1D), which resulted in 24 DC subclusters (Fig. 1E).

Within the cDC1 lineage, we identified cDC1 (Cl 2: *Xcr1, Cd24a, Itgae, Clec9a*), proliferating cDC1 (Cl 3: Prolif-cDC1: *Xcr1, Clec9a, Mki67, Stmn1*), and Pre-cDC1[15] (Cl 4: high expression of *Rab7b, Pbx1, Ifi205* in combination with low expression of cDC1 marker genes, including *Xcr1, Clec9a, Itgae*). Two Ccr7 clusters, Cl 1 and Cl 17, belonging to the cDC1 lineage were identified (based on the expression of *Cd81*, which is highly expressed in cDC1[52,74] in combination with XCR1 and CD103 ADT signal of the CITE-seq datasets) (Figs. 1E, G, H, Supplementary Fig. 2I, Supplementary Data 4, 5). Of these, Ccr7 Cxcl9 cDC1 (Cl 1: *Cxcl9, Cadm1, Il12b, Cd81*) was found close to the bulk of other cDC1s on the UMAP plot and had a lower expression of the maturation signature (*Ccr7, Il12b*), but highly expressed classical cDC1 markers, indicating that this cluster likely represented early stages of maturation. Contrastingly, Ccr7 cDC1 (Cl 17: *Ccr7, Fscn1, Laptm4b, Il12b*) clustered together with the other Ccr7 DCs and lacked the RNA expression of several classical cDC1 markers, while expressing the cDC1-specific gene *Cd81*. (Fig. 1E, G, H, Supplementary Fig. 2I, J, Supplementary Data 4, 5).

Within the pDC lineage, we identified pDC (Cl 9: *Siglech, Ccr9*) and pre-pDC (Cl 10: *Siglech, Cd7, Apod, Cd200r1*). Furthermore, we identified DC lineages including cDC2A (Cl 21: *Ltb, Ffar2, Pglyrp1, Ppp1r14a*)[2,3,28], cDC2B (Cl 23: *Gas5, F11r*), and DC3 (Cl 22: *Lyz2, Cd209a, Vim*)[2] (Fig. 1E, G, Supplementary Data 4). Importantly, besides lineage-defined subsets, we identified distinct DC states, such as DCs expressing an ISG signature (*Ifi206, Ifit3, Oasl2, Rsad2, Ifi207*), including ISG-DC (Cl 5), ISG-DC3 (Cl 6), and ISG-Pro-DC3 (Cl 7). Other DC states included Hypoxic-DC3 (Cl 13: *Hilpda, Vegfa, Egln3, Bnip3*), and proliferating cDC2 (Cl 24: *Mki67, Stmn1*) (Fig. 1E, G, Supplementary Data 4, 5). Importantly, we identified two additional clusters, Cl 12 and Cl 20, which we annotated as Cd83 DC (*Cd83, Nr4a1, Il1b*) and Mgl2 DC (*Mgl2, Fcgr2b, Clec4n, Il1r2, Mrc1*) respectively, since, based on classical markers, we were not able to assign them to a particular lineage, nor could we affirm that they represent DC states (Fig. 1E, G, Supplementary Data 4, 5). Cl 11, with a high expression of *Cd300c, Cd200r1*, and *Cd7*, was annotated as cDC2 progenitors (Supplementary Data 4). This cluster was observed between the cDC2/DC3 and pDC clusters on the UMAP plot, a feature commonly observed for cDC2 progenitors[16,24]. Moreover, a cluster of Pro-DC3 (Cl 8: *Ccl6, Ccl9, Cd209a, Lyz2*)[3] was identified. Finally, 7 subclusters of Ccr7 DCs, including those seemingly derived from Pre-cDC2A/cDC2A (Cl 14: Ccr7 Pre-cDC2A and Cl 16: Ccr7 cDC2A, respectively), mixed cDC2s (Cl 19: Ccr7 cDC2), and DC3 (Cl 15: Ccr7 DC3) were identified. One cluster (Cl 18) did not express cDC2, DC3, or cDC1 genes, and hence was annotated as Ccr7 DC, potentially encompassing DCs that have lost their ontogenic signature (Fig. 1E, G, Supplementary Fig. 2J, Supplementary Data 4, 5).

After initial annotation of the DC clusters primarily based on the expression of classical marker genes (Fig. 1), we independently validate these annotations in the following section using gene signatures

https://doi.org/10.1038/s41467-026-73721-w



obtained from fate-mapping studies that define distinct DC lineages (Fig. 2).

To further dissect the cDC2/DC3 states, we subsetted and re-clustered the cDC2A, cDC2B, DC3, ISG-DC, ISG-DC3, ISG-Pro-DC3, Pro-DC3, cDC2 Progenitors, Cd83 DC, Hypoxic-DC3, and Mgl2 DC (Fig. 1E, F). The cDC2A cluster split into three subclusters (Cl 21.1: cDC2A (1), Cl 21.2: cDC2A (2), and Cl 21.3: cDC2A (3)) with an overall shared gene backbone (*Ltb, Ffar2, Cd72*) but distinct DEGs (Supplementary Fig. 2H). Importantly, cDC2A and cDC2B clusters appeared to also share several genes compared to other DC clusters; however, upon zooming into them, it was evident that they possessed distinct gene signatures (Supplementary Fig. 2H) (further validation about the distinct nature of cDC2B from cDC2A is elaborated in the next section (Fig. 2)). Furthermore, the ISG-DC3 cluster split into two subclusters, Cl 6.1 ISG-DC3

Fig. 2 | Annotation of the TADC clusters within the mouse atlas was based on gene signatures from fate-mapping studies. A Violin plot showing the Pre-cDC1 gene signature score generated using the bone marrow (BM) dataset of Liu et al.[2], with the highest score detected within the Pre-cDC1 cluster. **B** Violin plot showing *Cd300c* gene expression across different TADC clusters. **C**–**F** Dotplots showing in the different mouse TADC atlas clusters (**C**) the averages of signature scores for pDC-like[3]/Noncanonical DC[21]/tDC[17]/SiglecH DC[28]; **D** the averages of signature scores for Pre-cDC2A[29]/ Pre-cDC2B[29]/Pro-DC3[2]/ Pre-DC2[2]; **E** *Tcf4* and *Klf4* average gene expression; **F** the average gene signature scores for pDC[2,17]. Blue squares highlight the target clusters. **G** Volcano plot showing DE genes between Pro-DC3 and Pre-cDC2B from the mouse TADC atlas. Red dots, DE genes with adjusted p-value < 0.05, $\log_2$(Fold change) > 0.5; Calculation of the p-values was performed with two-sided Wilcoxon Rank Sum test, followed by correction for multiple testing by the Bonferroni method. **H** Heatmap showing average scaled gene signature scores of splenic Tbet$^+$/Tbet$^-$ DCs[28], ESAM$^+$ cDC2[3], DC2[2], DC3[2,3], and cDC2b[3]. **I** Volcano plots showing DE genes between DC3 and cDC2B from the mouse TADC atlas. Red dots, DE genes with adjusted p-value < 0.5, $\log_2$(Fold change) > 0.5. Calculation of the p-values was performed with two-sided Wilcoxon Rank Sum test, followed by correction for multiple testing by the Bonferroni method. **J**, **K** Dotplots showing average expression across the different TADC clusters of **I** interferon-stimulated genes (ISG); **J** hypoxia-associated genes.

(1) and Cl 6.2 ISG-DC3 (2), with Cl 6.1 located closer to the DC3 cluster in the UMAP, potentially highlighting that it is in the early stage of acquiring an ISG response (Fig. 1F, G). Contrastingly, Cl 6.2 clustered further away and closer to other ISG-DCs, suggesting a more advanced ISG phenotype (Fig. 1F, G). ISG-DC split into three subclusters: Cl 5.1 ISG-cDC2 seemed to belong to the cDC2 lineage; Cl 5.2 ISG-DC could not be associated with a specific DC lineage and likely encompassed different cDC2/DC3 subsets; and Cl 5.3 Ccr7 ISG-DC showed a low degree of *Ccr7* and *Ccl5* expression, indicating a potential trajectory from the ISG to the Ccr7 states (Fig. 1F, G). The cDC2 progenitors cluster (Cl 11) further split into three subclusters, including the previously defined pDC-like cells[2,3,16,21,24] (Cl 11.1: *Apod, Cd7, Cd200r1, Runx1*), Pre-cDC2A[29] (Cl 11.2: *Adam23, Ch1, Alox5, Cyp7b1*), and Pre-cDC2B[29] (Cl 11.3: *Gria3, Cd7, Pdlim1, Lmo1*) (Fig. 1E–G, Supplementary Data 4, 5). Each DC subset/state was present in varying percentages within the models for different cancer types, highlighting cancer-type specific differences within the TADC subsets (Fig. 1I, Supplementary Fig. 3A–C). Since several datasets in the mouse atlas did not have replicates, statistical analyses were limited to DCs from GL261, MC38, KPC, and MMTV-PyMT tumors (Supplementary Fig. 3A, Supplementary fig. 3C). As such, the atlas serves primarily as a qualitative resource for mapping cellular diversity, and observed variations in abundance should be interpreted as indicative trends rather than exact quantitative assessment. Nonetheless, when comparing the distribution of DC clusters across cancer models, we noted that pDCs appeared more prevalent in breast tumor models such as MMTV-PyMT, 4T1, and EMT6, but not EO771 (Fig. 1I, Supplementary Fig. 3A–C). Mgl2 DC were more abundant in KPC, KPM, and LLC tumors (Fig. 1I, Supplementary Fig. 3A–C). Interestingly, while LLC and LLC-OVA lung carcinomas differ in antigenicity, they displayed a largely comparable composition of DC clusters (Fig. 1I).

We next compared the relative abundance of DC subsets between subcutaneous, orthotopic and spontaneous/inducible mouse models, and found that cDC1s, Ccr7 DC, Mgl2 DC, and Cd83 DC were more abundant in spontaneous/inducible compared to subcutaneous models, with Mgl2 DC and Cd83 DC also showing a significant increase compared to orthotopic models (Supplementary Fig. 3D). Conversely, subcutaneous models contained a higher proportion of DC precursors, while ISG-DCs were enriched in both subcutaneous and orthotopic models (Supplementary Fig. 3D).

Overall, we present a scRNA-seq TADC atlas from 14 preclinical mouse cancer models consisting of 31 DC subsets and states with distinct gene signatures.

TADC cluster annotations within the mouse atlas are validated using gene signatures from fate-mapping studies on DC ontogeny

While defining cDC1s, pDCs, and Ccr7 DCs is straightforward owing to their distinct gene signatures, distinguishing the different cDC2 and DC3 subsets and states currently remains challenging. To date, multiple (single-cell/bulk) RNA-seq studies have been conducted to better understand the different DC subsets in homeostasis and inflammation[2,3,16,21,23,24,28–30,52,75]. Out of these, we chose studies that determined DC lineages via fate mapping techniques as the backbone of our analysis and DC annotation[2,3,17,28,29]. By integrating these complementary datasets into a unified platform, we aimed to create a more comprehensive framework for identifying DC subsets and states in cancer. Using RNA-seq gene expression matrices from these studies, we generated signature scores for the different DC subsets (Supplementary Data 6)[2,3,17,21,28,29]. The top upregulated genes in the relevant DC populations from each study were used to calculate additive signature scores in the TADC atlas, avoiding the potential bias from inspecting individual gene expression, which could be obscured by technical noise. As such, we use the Pre-cDC1 score based on the BM scRNA-seq dataset of Liu et al.[2] to validate the cluster we annotated as Pre-cDC1 (Fig. 2A). We observed that this score was high in nearly all the clusters, but higher within the cDC1 lineage. Among the cDC1 lineage, it was marginally the highest in the Pre-cDC1 cluster, validating our initial annotation (Fig. 2A). As an alternate method to validate the progenitor state of this cluster, RNA velocity analysis performed using Velocyto[76] showed a directional flow from pre-cDC1 to cDC1, strengthening this annotation (Supplementary Fig. 4A, B). To validate the other DC progenitor clusters, we examined the expression of *Cd300c* and found that besides pDC and Pre-pDC clusters, *Cd300c* was only expressed in three other clusters, which we annotated as pDC-like, Pre-cDC2A, and Pre-cDC2B (Fig. 2B). *Cd300c* is selectively expressed in pDC, pre-pDC, and pre-cDC2s but not in Pre-cDC1 or Pro-DC3[3,17,25], which is in line with our observations (Fig. 2B). The pDC-like cells signature score from the Rodrigues et al.[3] dataset scored highest within the Pre-cDC2B cluster, and second highest in the pDC-like cluster (Fig. 2C). Given a certain degree of overlap between the Rodrigues et al. pDC-like cells with noncanonical DCs[21], tDCs[16,17], and SiglecH DCs[28], we assessed the gene signatures of each of these populations within our clusters (Fig. 2C). While all these scores showed a certain degree of expression in the pDC-like cells, they scored higher in other progenitor clusters. For instance, the noncanonical DC[21], tDC-hi[17], and SiglecH DC[28] scores were the highest in our Pre-cDC2A cluster, while the tDC-lo[17] score was highest in our Pre-cDC2B and Pro-DC3 clusters (Fig. 2C). Given the inconsistency of these scores seen within the progenitor populations in the atlas, we first aimed to correctly define the pre-cDC2A and pre-cDC2B clusters. Minutti et al. have elegantly described BM imprinting of Pre-cDC2 to the cDC2A and cDC2B lineages using fate-mapping studies[29]. Using their Pre-cDC2A and Pre-cDC2B-derived bulk RNA-seq gene signature scores[29], we validated the (Ccr7) Pre-cDC2A and Pre-cDC2B in our TADC atlas (Fig. 2D). Importantly, Minutti et al. excluded Ly6D$^+$ cells while sorting Pre-cDC2A and Pre-cDC2B, whereby precursors of pDC and tDC were excluded[29]. Additionally, Minutti et al. have shown that upon comparison of bulk-RNA sequenced tDC, CD8$^+$ pre-cDC2 (putative pre-cDC2A) and CD8$^-$ pre-cDC2 (putative pre-cDC2B), the gene expression profiles of each of these cell types were distinct[29], highlighting that these scores are the most accurate to define Pre-cDC2A and Pre-cDC2B. *Tcf4* (a transcription factor expressed in cDC2 progenitors[3,25]) expression was detected within the Pre-cDC2A and Pre-cDC2B clusters (Fig. 2E). The transcription factor *Klf4* has been shown to play a key role in cDC2B development[25,77] and was predominantly expressed in the

Pre-cDC2B cluster (Fig. 2E). Hence, based on the gene signature scores from Minutti et al., in combination with the *Tcf4* and *Klf4* expression, we could validate Pre-cDC2A and Pre-cDC2B in our atlas (Fig. 2D, E). Using this as an exclusion strategy, the cluster that scored second highest for the pDC-like score from Rodrigues et al.[3], as well as highest for the pDC score from all the non-pDC lineage cells, was annotated as pDC-like cells, highlighting their intermediate transcriptional phenotype between cDC2 and pDC (Fig. 2C, F). As for the variable expression of the noncanonical DC, tDC, and SiglecH DC scores in the Pre-cDC2A and Pre-cDC2B clusters rather than in the pDC-like cluster, we hypothesize that since these studies did not further dissect the pre-cDC2 (which overlap to a certain extent with pDC-like cells), their analysis could contain some Pre-cDC2A/B (Fig. 2C).

DC3 and cDC2B lineages are known to showcase overlapping gene signatures, including multiple monocyte/macrophage-associated genes[2,3,29]. However, using fate-mapping studies, it has recently been defined that DC3 and cDC2B originate from different immediate progenitors, namely the Pro-DC3 and Pre-cDC2B, respectively[3]. We validated Pro-DC3 in our TADC atlas using the Pro-DC3 and Pro-DC3 versus Pre-DC2 scores based on the splenic dataset of Liu et al.[2] (Fig. 2D). A comparison between Pro-DC3 and Pre-cDC2B showed upregulation of several genes within Pro-DC3 including *Cyba, Ms4a4c, Ccl6, Cd74, Lyz2, Tgfbi, H2-Ab1, H2-Aa*, and *H2-Eb1*, while Pre-cDC2B showed higher expression of *Cd7, Aff3, Bst2, Cd300c, Tcf4, Gria3 and Siglech* among others (Fig. 2G). Interestingly, most of these genes aligned with those previously reported as DEGs in Pro-DC3 and Pre-DC2 in the BM, blood, and spleen[2].

To validate the lineage identity of cDC2A, cDC2B, and DC3 clusters in our atlas, we first assessed the T-bet$^+$ DC (defining cDC2A) versus T-bet$^-$ DC score from Brown et al.[28], wherein the three cDC2A clusters, as well as Ccr7 Pre-cDC2A, Ccr7 cDC2A, scored high, enabling us to validate them as bonafide cDC2A (Fig. 2H, Supplementary Fig. 4C). A similar score pattern was observed with the ESAM$^+$ cDC2 score[3]. Evidently, the T-bet$^-$ versus T-bet$^+$ DC score was highest in the non-cDC2A clusters (Fig. 2H, Supplementary Fig. 4C). Among the T-bet$^-$ clusters, DC3 identity was confirmed using DC3 scores[2,3]. Importantly, the cDC2B cluster in the atlas was validated based on its relatively high splenic DC2 and DC3 scores from Liu et al. (Fig. 2H, Supplementary Fig. 4C). Liu et al. performed scRNA-seq on Lin$^-$CD45$^+$CD16/32$^{lo\text{-}hi}$Flt3$^+$ splenocytes and clustered the cells into DC2, DC3 and their progenitors[2]. As the DCs were not further segregated into cDC2A and cDC2B, it is plausible that some cDC2B are included within the DC2 and DC3 owing to their shared similarities to these lineages. Hence, theoretically both the Liu et al. DC2 and DC3 scores would also be present within the cDC2B cluster of the TADC atlas. Indeed, the Liu et al. DC2 and DC3-specific scores were also high in the cDC2B cluster (Fig. 2H, Supplementary Fig. 4C). A comparison of the DEGs between DC3 and cDC2B revealed that cDC2B showed higher expression of genes, including *Cd72, Id2, Cst3, Gas5*, and MHC-associated genes (*B2m, H2-Q7, H2-Oa*), while DC3 upregulate genes previously observed by Liu et al. (upon comparing DC2 with DC3) including *Ccl9, Cd209a, Vim*, and *Ccr2* (Fig. 2I, Supplementary Fig. 4F). Overall, using the T-bet$^+$ versus T-bet$^-$ DC scores[28] and ESAM$^+$ cDC2[3] scores, we validated cDC2A clusters (Fig. 2H, Supplementary Fig. 4C). Within the non-cDC2A clusters, the DC3 were confirmed using the DC3 scores[2,3]. Finally, based on the rather broad DC2 and DC3 scores from Liu et al.[2] both of which we consider to encompass some cDC2Bs, we identified the cDC2B cluster (Fig. 2H). Additionally, we provide a list of genes shared between our mouse TADC subsets and the major DC subsets from the public murine RNA-seq studies we used to generate signature scores (Supplementary Fig. 4G).

Several DC states, including ISG-DC, ISG-cDC2, ISG-Pro-DC3, and ISG-DC3, were validated based on their general DC, cDC2-like, Pro-DC3-like, and DC3-like signatures, respectively, in addition to the expression of the inflammatory/interferon-stimulated genes, including *Irf7, Ifit1, Ifit2* (Fig. 2H, J, Supplementary Fig. 4C). Additionally, Cd83 DC, and Mgl2 DC were annotated as DC without any lineage specification, as these clusters could not be restricted to a particular lineage due to their broad expression of cDC2, (pre-) cDC2B, and (Pro-) DC3 scores (Fig. 2C, D, H, Supplementary Fig. 4C). These clusters are probably a mix of ontogenic lineages, and due to their overwhelming gene signatures, delineating the underlying lineages is challenging. The differential expression of *Mgl2* (gene encoding CD301b) and *Cd83* was used to annotate the Mgl2 DC and Cd83 DC, respectively (Supplementary Fig. 4D, E). Hypoxic-DC3 annotation was validated using the DC3 scores in combination with high expression of hypoxia-associated genes, including *Hilpda, Vegfa, Hmox1, Slc2a1, Egln3*, and *Hif1a* (Fig. 2H, K).

RNA velocity analysis revealed a directional flow from several Ccr7 lineage-associated clusters ultimately ending in the lineage-negative Ccr7 DC cluster, highlighting that as DCs mature, they lose their lineage signatures and get unified under Ccr7 DC state (Supplementary Fig. 4A). A directional flow was also observed from pre-cDC1, cDC1 to the Ccr7 cDC1, ultimately ending in the Ccr7 DC (Supplementary Fig. 4B).

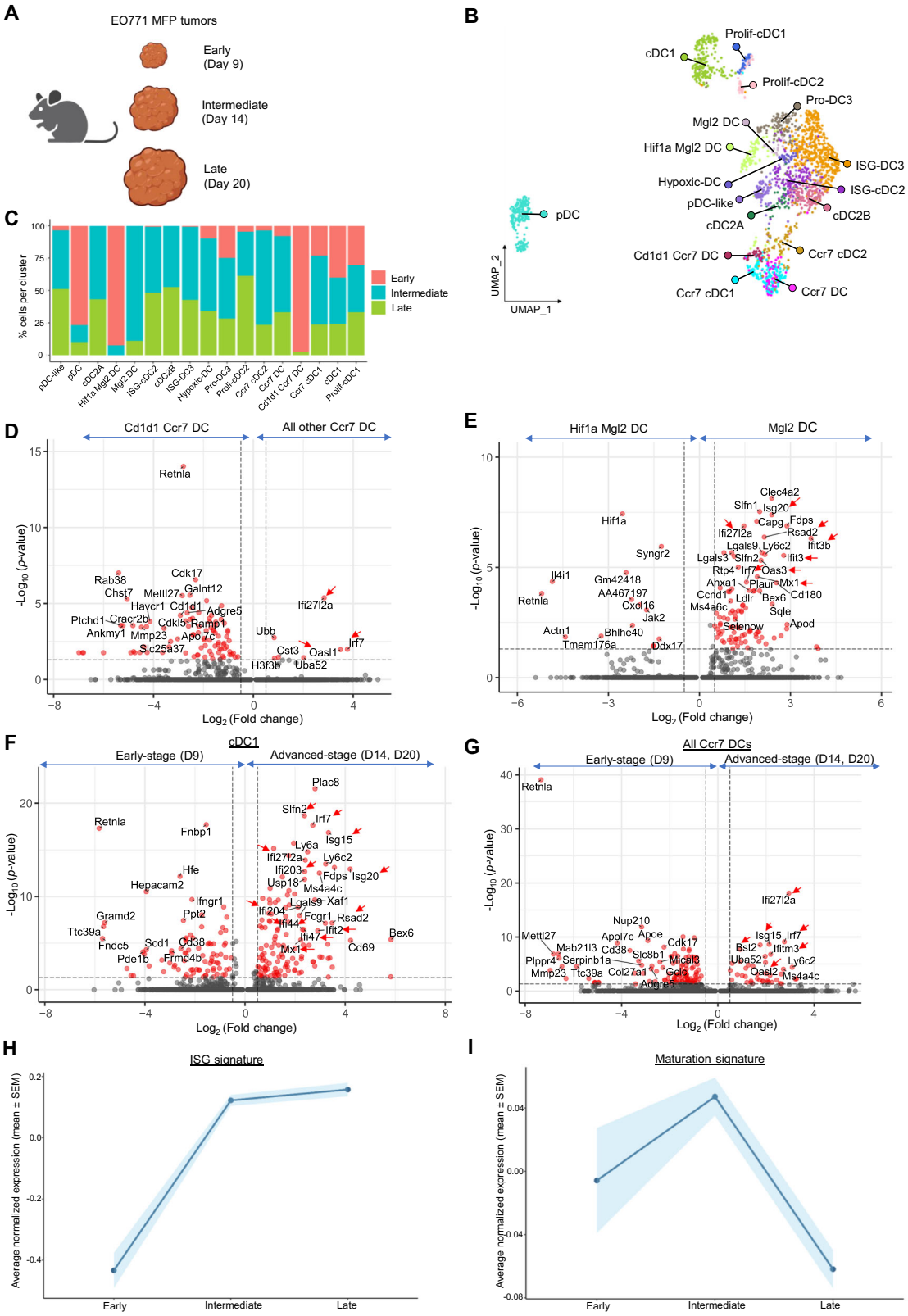
To confirm that our approach did not lead to overclustering, we applied the sc-SHC algorithm[78] to assess the statistical significance of the final TADC clustering (Supplementary Fig. 5). This method hierarchically clusters the centers of the provided clusters and performs recursive statistical testing at each node of the resulting hierarchical tree. All TADC subsets showed significant separation, except ISG-DC and ISG-cDC2 (Supplementary Fig. 5A, B). As ISG-cDC2 showed a specific cDC2 lineage signature score (Fig. 2H), this population was retained as a separate cluster.

Collectively, using gene signature scores from the presiding DC lineage tracing studies, we validated the cDC1, cDC2A, cDC2B, and DC3 lineages that we previously identified using marker genes, in addition to their respective progenitors and activation states within our mouse pan-cancer TADC atlas.

TADCs acquire an inflammatory profile with tumor progression

Several changes occur within the immune and non-immune compartments of the tumor microenvironment (TME) with tumor progression[79]. To evaluate how tumor progression impacted the TADC subsets and states, we performed kinetic scRNA-seq on EO771 tumors at 9 (early), 14 (intermediate), and 20 (late) days post tumor inoculation (Fig. 3A, Supplementary Fig 6A). Most TADC clusters identified in the atlas could be mapped on the kinetic dataset (Fig. 3B, Supplementary Data 7). While the DC heterogeneity did not vary much in intermediate compared to late-stage tumors, two new DC clusters, namely the Hif1a Mgl2 DC and Cd1d1 Ccr7 DC, were identified and were predominantly present in early-stage tumors (Fig. 3B, C, Supplementary Fig. 6B, C). Additionally, *Cd1d1* expression was majorly restricted to TADCs from early tumors (Supplementary Fig. 6F). Upon assessing the DEG between Cd1d1 Ccr7 DC and all other CCR7 DC clusters, we found that the latter upregulated ISGs, including *Ifi27l2a, Oasl1*, and *Irf7* (Fig. 3D). Similarly, a comparison of Hif1a Mgl2 DC (predominant in early tumors) with Mgl2 DC (exclusively found in intermediate and late-stage tumors) revealed upregulation of ISGs in the latter, including *Isg20, Ifit3b, Ifi27l2a, Rsad2, Ifit3, Oas3*, and *Irf7* (Fig. 3E). Interestingly, a consistent pattern of ISG upregulation was also observed in advanced-stage (D14, D20) cDC1, Ccr7 DC, Pro-DC3, and pDCs compared to their early-stage (D9) counterparts (Fig. 3F-G, Supplementary Fig. 6D, E). Upon assessing the expression of a larger panel of 30 ISGs in all TADCs from EO771 tumors across tumor progression, we observed that, as tumors progressed from early to late stages, the ISG signature was amplified (Fig. 3H). Using a panel of 32 DC activation genes, we observed an initial increase in DC activation genes[4,18,80,81] in the intermediate stage, followed by a sharp decrease in the late stage (Fig. 3I).

https://doi.org/10.1038/s41467-026-73721-w



To evaluate whether these changes were model-specific or could be extended to other models, we performed kinetic analysis of TADCs in four additional datasets, including inducible KP (lung cancer), spontaneous KPCY (PDAC)[82], inducible GBM (CAG-LSL-EGFR^{WT};InkΔ2/3$^{-/-}$;PTEN^{2lox})[83], and subcutaneous B16F10 (melanoma) tumors[84] (Supplementary Fig. 7, Supplementary Data 8–11). The Cd1d1 Ccr7 DC population could be identified in KP, KPCY, and GBM tumors; however, potentially due to the limited number of cells, this cluster could not be defined within B16F10 tumors (Supplementary Fig. 7A, B, Supplementary Fig. 7E, F, Supplementary Fig. 7I, J, Supplementary Fig. 7M). Furthermore, in KP and GBM tumors, Cd1d1 Ccr7 DCs were most abundant at early stages, and *Cd1d1* expression peaked early in tumor development, consistent with our observations in EO771 (Supplementary Fig. 6C, Supplementary Fig. 6F, Supplementary Fig. 7C, D,

Fig. 3 | TADCs augment interferon-stimulated inflammatory gene signature with tumor progression. A Schematic overview of the scRNA-seq datasets used to generate the mouse EO771 kinetic dataset, including EO771 tumors from days 9 (early, $n = 1$ from 12 pooled tumors), 14 (intermediate, $n = 1$ from 4 pooled tumors), and 20 (late, $n = 1$ from 4 pooled tumors) post-inoculation. Created in BioRender. Laoui, D. (2026) https://BioRender.com/zz1q7gs. **B** UMAP plot of 1960 TADC from early, intermediate, and late-stage EO771 tumors, clustered into subsets and states. **C** Bar graph showing the percentage of DC originating from early, intermediate, and late-stage tumors per cluster. Volcano plots showing DE genes between **D** Cd1d1 Ccr7 DC and all other Ccr7 DC; **E** Hif1a Mgl2 DC and Mgl2 DC; **F** cDC1 from early-stage (D9) and advanced-stage (D14, D20) tumors; **G** All Ccr7 DC from early-stage (D9) and advanced-stage (D14, D20) tumors. Red dots, DE genes with adjusted p-value < 0.05, $\log_2$(Fold change) > 0.5. Calculation of the p-values was performed with two-sided Wilcoxon Rank Sum test, followed by correction for multiple testing using the Bonferroni method. The red arrows highlight examples of interferon-stimulated genes. **H–I** Ribbon plots highlighting changes of average scaled normalized expression (mean ± standard error of mean) within all TADCs across early, intermediate and late-stage EO771 tumors of **H** selected Interferon-stimulated and inflammation-associated genes (*Adar, Bst2, Ccl5, Cxcl10, Eif2ak2, Ifi204, Ifi209, Ifi211, Ifi35, Ifi44, Ifih1, Ifit2, Ifit3, Ifit3b, Isg20, Irf9, Nmi, Oasl1, Oas1c, Oas1g, Rsad1, Slfn5, Tnfsf10, Trim5, Trim25, Zc3hav1, Oas2, Ifitm10*); **I** selected DC-activation and maturation genes (*Ccr7, Fscn1, Ccl5, Marcksl1, Mir155hg, Stat4, Traf1, Gadd45b, Rel, Relb, Cd274, Pdcd1lg2, Cd200, Cd63, Tspan3, Serpinb6b, Serpinb9, Cflar, Il15, Il15ra, Cd80, Cd83, Cd40, Cd86, Tnfrsf4, Adcy6, Socs2, Cacnb3, Sct, Il4i1, Glipr2, Tmem176a*). The solid line represents the average expression across genes, and the shaded ribbon indicates the standard error of the mean (SEM) calculated across genes.

Supplementary Fig. 7K, L). In contrast, KPCY was the only model in which Cd1d1 Ccr7 DCs were predominant in late-stage tumors (Supplementary Fig. 7G, H).

Next, we analyzed whether different TADC types displayed distinct patterns of ISG and maturation signature expression. For this, we broadly categorized the clusters as cDC1, cDC2_DC3, pDC, or Ccr7 DCs. In EO771, all populations exhibited an increase in ISG signature (Supplementary Fig. 8A). Similarly, in KP, KPCY, and B16F10, a modest increase was observed in the ISG signature within cDC2_DC3, while the other population showed varied effects (Supplementary Fig. 8B, C, Supplementary Fig. 8E). Importantly, GBM showed a sharp decrease in ISG signature at the intermediate tumor stage, followed by an increase at the late-tumor stage (Supplementary Fig. 8D). Similar to EO771, DC2_DC3 from all models, except KP, showed a trend for decrease in the maturation signature (Supplementary Fig. 8F–J). In all, we show that as tumors progress from early to late stages, the interferon-stimulated inflammatory gene signature of cDC2 and DC3 intensifies and maturation signature diminishes, although in varying levels depending on the model, with EO771 seemingly being highly inflammatory.

Cancer drives reprogramming of DC subsets and states

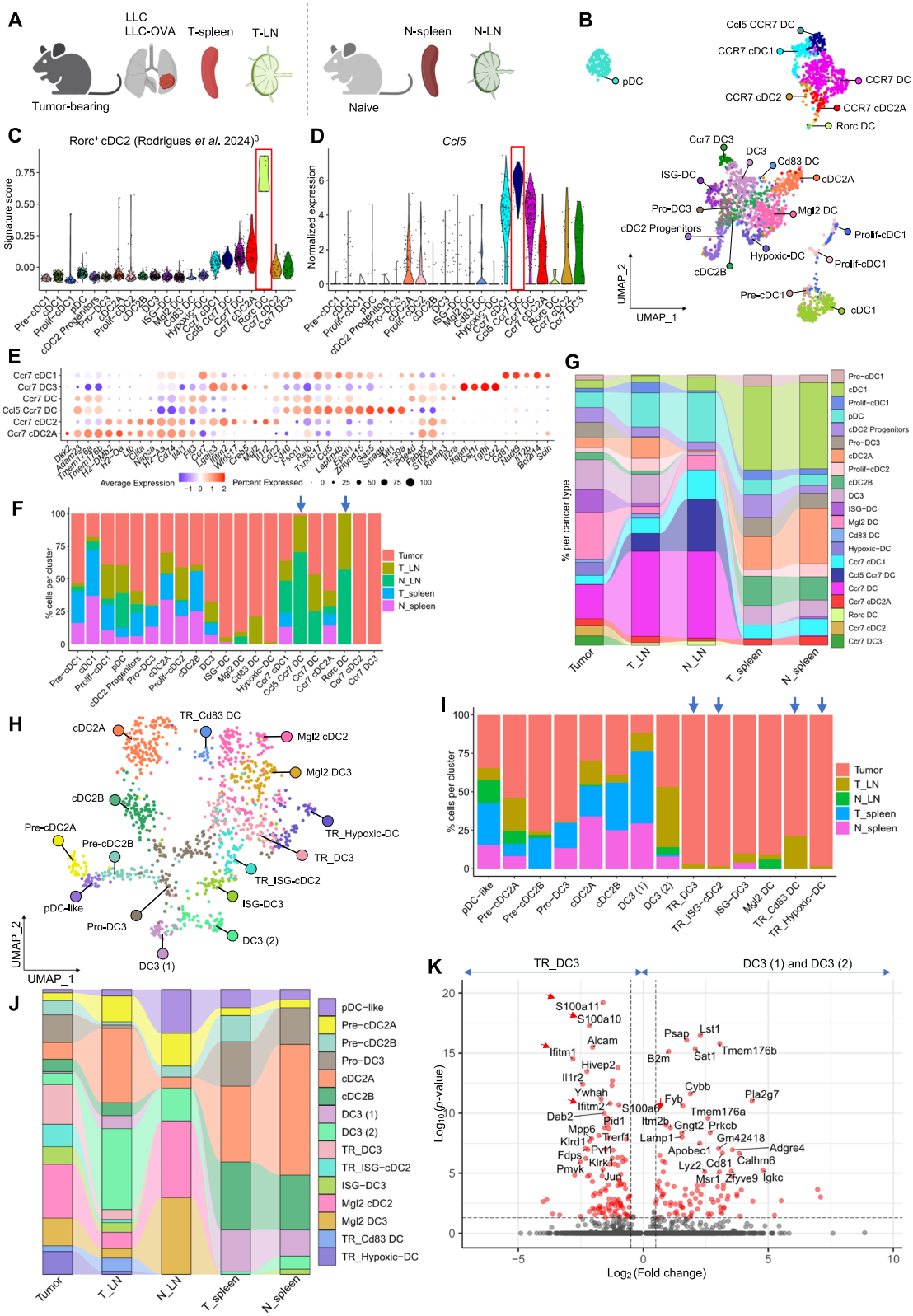
Tumors are known to induce systemic immune changes[85,86]. To investigate the impact of tumors on the DC compartment of the spleen and LN, we generated an integrated organ scRNA-seq dataset composed of LLC and LLC-OVA tumors from our TADC atlas along with spleens and LN from LLC-OVA bearing (T-spleen and T-LN, respectively) and naïve mice (N-spleen and N-LN, respectively) (Fig. 4A, B). We first identified all the broad DC clusters observed in the mouse TADC atlas based on their correlating gene signatures (Fig. 4B, Supplementary Fig. 9F, Supplementary Data 12). Interestingly, two additional DC clusters were identified, namely the Rorc DCs, identified using the Rorc$^+$ cDC2 score by Rodrigues et al.[3] (Supplementary Data 6, Fig. 4B, C), and Ccl5 Ccr7 DCs, which were marked by a high expression of *Ccl5, Epsti1, Zmynd15, Mt1, Smagp* (Fig. 4B, D, E, Supplementary Data 12). *Rorc* (encoding RORγt) expressing DCs have been described in mice and humans, and are attributed with a range of functions including LN migration, CD4$^+$ T cell stimulation, and Treg-mediated tolerance induction[87–89]. Potentially owing to their LN migratory capacity[88], these cells clustered close to the Ccr7 DCs (Fig. 4B). Notably, in our integrated organ dataset, Rorc DCs were found exclusively within LNs from tumor-bearing and naïve mice (Fig. 4F, G). Rorc DCs have also been reported to form a preferential niche in the LNs[90]. We found Ccl5 Ccr7 DCs predominantly in the N-LN and T-LN, but negligible within tumors (Fig. 4F, G). Rawat et al. reported a murine LN scRNAseq dataset with a similar Ccl5$^+$ Ccr7$^+$ DC subset (migDC1, also upregulating *Epsti1, Zmynd15, Mt1, Smagp*), which was shown to guide antigen-loaded monocytes to the draining LN[91]. Owing to their predominant presence within LN of naïve and tumor-bearing mice, we hypothesized that the Ccl5 Ccr7 DC could represent LN resident and potentially tolerogenic DCs (Fig. 4B, F–G). To validate this further, we aligned our dataset with that from Rennen et al.[80] wherein they used lipid nanoparticles (LNP) as a tool to investigate cDC1 maturation. Their dataset comprised cDC1s sorted from the spleen during steady state and 2 or 8 hours post administration of: (1) CpG-encapsulating LNP (CpG-LNP), (2) PolyI:C encapsulating LNP (pIC LNP), (3) PolyI:C only, and (4) empty LNP (eLNP) (Supplementary Fig. 9A–C). Rennen et al. observed that cDC1s exhibit different maturation states ranging from immature, early mature, to late mature states, and depending on the stimulus, could mature in an immunogenic or homeostatic manner (Supplementary Fig. 9A–C). By projecting our integrated organ dataset on the LNP reference UMAP, we show that our Ccl5 Ccr7 DCs align with the "late mature" cDC1s from steady state, suggesting that this cluster could be homeostatic (tolerogenic) mature DCs (Supplementary Fig. 9D, E).

To better appreciate the cDC2/DC3 heterogeneity in the secondary lymphoid organs of tumor-bearing and naïve mice, we subsetted and re-clustered the cDC2 Progenitors, Pro-DC3, cDC2A, cDC2B, DC3, ISG-DC, Hypoxic-DC, Cd83 DC, and Mgl2 DC from the integrated organ dataset (Fig. 4B, H). Herein, all the cDC2 and DC3 subsets/states and their progenitors identified within the TADC atlas could be retrieved (Fig. 4H, Supplementary Fig. 9F, Supplementary Data 13). Importantly, the DC3 split into three clusters, with one predominantly found within the tumor and lymphoid organs of tumor-bearing mice, which we termed tumor-reprogrammed (TR) DC3 (Fig. 4H–J). To determine potential differences between the TR_DC3 and non-TR DC3 clusters, we evaluated the DEGs and observed upregulation of ISGs, including *Ifitm1, Ifitm2*, as well as inflammation-associated genes such as *S100a6, S100a10*, and *S100a11* within the TR_DC3 (Fig. 4K). Additionally, we identified TR_ISG-cDC2, TR_Cd83 DC and TR_Hypoxic-DCs that were also predominantly present in the tumor and to a lower extent within the tumor-draining LN (Fig. 4H–J), indicating tumor-mediated reprogramming in these clusters. In conclusion, we demonstrate that the presence of the tumor induces (inflammatory) reprogramming in the DC compartment, giving rise to DCs that are exclusively present in the tumor or lymphoid organs from tumor-bearing mice and are absent within naïve lymphoid organs.

Single-cell RNA sequencing reveals the TADC heterogeneity in 10 human cancer types

To unravel the human TADC heterogeneity, we generated a human pan-cancer TADC atlas using publicly available scRNA-seq datasets. The atlas comprises 10 cancer types including breast carcinoma (BC), colorectal carcinoma (CRC), glioblastoma (GBM), hepatocellular carcinoma (HCC), head and neck squamous cell carcinoma (HNSCC), lung carcinoma (LC), melanoma, ovarian carcinoma (OC), pancreatic ductal adenocarcinoma (PDAC), and renal cell carcinoma/Wilms tumors (RCC)[28,32,33,35,92–100] (Fig. 5A). The available patient details of interest are listed in Supplementary Data 14. The generation of the human TADC atlas followed the same analysis pipeline as for the mouse atlas, wherein we first generated the complete human tumor MNP atlas,

https://doi.org/10.1038/s41467-026-73721-w



integrating 61,432 cells (Fig. 5B, Supplementary Fig. 10A–C). Within the MNP atlas, we mapped monocytes (Mono), macrophages (Mac), proliferating monocytes and macrophages (Prolif-Mono_Mac), cDC1, cDC2, CCR7 DC, proliferating DC (Prolif-DC), and pDC based on DEGs (Fig. 5B, Supplementary Fig. 10B, Supplementary Data 15). Four monocyte subclusters were identified, including non-classical monocytes (NonClass-Mono: *FCGR3A, LST1, CDKN1C*), classical monocytes (Class-Mono: *FCN1, CFP, S100A8/9*), IL1B Mono (*IL1B, SOD2, TIMP1, EREG, CCL20*), and heat shock protein (HSP)-Mono (*HSP90AA1, HSPA1A, HSPA1B*) (Supplementary Fig. 10B, D). We annotated 13 macrophage clusters including Immature-Mac (with intermediate gene signatures between monocytes and macrophages), lipid-associated macrophages (Lipid-Mac: *GPNMB, CTSB, APOC1, TREM2*), FOLR2 Mac (*FOLR2, SEPP1, SLC40A1, STAB1*), TGFBI Mac (*TGFBI, SPP1, STAB1*), Hypoxic-Mac (*BNIP3, HILPDA, ERO1A*), IER3 Mac (*IER3, NR4A2, FOSB*), HSP-Mac (*HSPA1A, HSPA1B, HSPH1, HSPB1*), CXCL9 Mac (*CXCL9,*

Fig. 4 | Cancer-mediated DC reprogramming occurs in lymphoid organs. **A** Schematic overview of the scRNA-seq datasets used to generate the mouse integrated organ dataset encompassing LLC tumors ($n = 2$ with 3 pooled tumors each)[43,58], LLC-OVA tumors, spleens, LN from LLC-OVA tumor-bearing ($n = 1$ with 3 pooled organs each) and naïve spleen and naïve LN ($n = 1$ with 3 pooled organs each). Created in BioRender. Laoui, D. (2026) https://BioRender.com/qktqscc. **B** UMAP plot of 2287 DCs from the tumors, spleens, and LNs, clustered into 21 subsets and states. **C**, **D** Violin plots showing the expression of **C** gene signature score for Rorc$^+$ cDC2 generated using the splenic scRNA-seq dataset from Rodrigues et al.[3], with the highest score detected within our Rorc DC cluster; **D** *Ccl5* gene expression. **E** Dotplot showing the average expression of genes specific for the different Ccr7-expressing DC clusters from (**B**). **F** Bar graph showing the percentage of DCs originating from the different organ samples per DC cluster from (**B**). **G** Alluvial plot, showing the relative frequency of the TADC subsets and states from (**B**) across the different organ samples. **H** UMAP plot of 1,080 cDC2/DC3 obtained by subsetting and reclustering the cDC2 Progenitors, cDC2A, cDC2B, Pro-DC3, DC3, ISG-DC, Hypoxic-DC, Mgl2 DC, and Cd83 DC clusters of the dataset from (**B**), resulting in 15 subclusters. **I** Bar graph showing the percentage of cDC2/DC3 subclusters originating from the different samples per DC cluster from (**H**). **J** Alluvial plot, showing the relative frequency of the DC2/DC3 subsets and states from (**H**) across the different organ samples; **K** Volcano plots showing DE genes between TR_DC3 versus DC3 (1) and DC3 (2). Red arrows indicate ISG and other inflammatory genes. Red dots, DE genes with adjusted p-value < 0.05, $\log_2$(Fold change) > 0.5. Calculation of the p-values was performed with two-sided Wilcoxon Rank Sum test, followed by correction for multiple testing using the Bonferroni method. T-spleen, spleen from tumor-bearing mice; T-LN, lymph nodes from tumor-bearing mice; N-spleen, spleen from naïve mice; N-LN, lymph nodes from naïve mice; Prolif, proliferating; ISG, interferon stimulated genes; TR, tumor-reprogrammed.

CXCL10, GBP1), ISG-Mac (*ISG15, IFIT1, IFIT3*), MT1G Mac (*MT1G, MT1X, MT2A*), CX3CR1 Mac (*CX3CR1, C3, TREM2*), FABP4 Mac (*FABP4, MARCO, MCEMP1*), and CTSK Mac (*CTSK, ATP6V0D2, TNFRSF11A, SIGLEC15*) (Supplementary Fig. 10B, D). Evidently, the different MNP clusters showed variations in abundance across the different cancer types (Supplementary Fig. 10E, F).

The DC compartment was annotated using elaborate gene signatures, including critical DC genes such as *FLT3, CD1C* (specific to human cDC2s), *XCR1, CCR7*, and *CLEC4C* (specific to human pDCs) (Fig. 5C, Supplementary Fig. 10D). To perform a deeper characterization of the DC heterogeneity, the DC clusters (cDC1, cDC2, Prolif-DC, pDC, and CCR7 DC) from the MNP atlas were subsetted and re-clustered (Fig. 5B, Supplementary Fig. 11A–C, Supplementary Fig. 11F, Supplementary Data 16). cDC2 was the most predominant DC type within the human TADC atlas (Supplementary Fig. 11D, E). To fine-map the heterogeneity of cDC2s, pDC-like, and CCR7 DCs, we subsetted and re-clustered these clusters (Fig. 5D, Supplementary Fig. 11B, Supplementary Fig. 11G).

Clustering of the human TADC atlas was validated using the scSHC significance analysis[78]. All cDC2/DC3 clusters showed significant separation. Two CCR7 DC clusters did not pass significance testing (Supplementary Fig. 12D); however, CCR7 cDC1 was retained as a distinct cluster due to its clear cDC1 gene signature despite the low cell number (Supplementary Figs. 11H, 12D).

While several counterparts of mouse TADC clusters could be identified in human cancers, some clusters were identified in only one species (Fig. 1E, F and Fig. 5D). Several DC progenitor clusters were identified, including pDC-like cells (*AXL, SIGLEC6*)[24], Pre-cDC2B (*ENHO, CFP, SELL, SDPR*), and Pro-DC3 (*FCN1, S100A9, VCAN, SELL, ENHO*) (Fig. 5D, F). Interestingly, SCENIC analysis revealed that the Pro-DC3 cluster exhibited high predicted regulon activity of the CEBPB, CEBPD, CEBPA, RARG, and NFE2 regulons, mirroring the increased activity observed in the mouse Pro-DC3 (Supplementary Fig. 12E, F). Similarly, human and mouse Pre-cDC2Bs both showed elevated activity of the MEF2C, FLI1, and JUN regulons (Supplementary Fig. 12E, F), further supporting our annotation. The Pre-cDC2A cluster found in mouse tumors was not identified in human tumors. Additionally, several different lineages of DC subsets were identified, including cDC2A (*CD1A, LTB, IL22RA2*), cDC2B (*CLEC10A, AREG, IL1R2*), and DC3 (*FCGR3A, C1QA, C1QC, CD14*) (Fig. 5D, F, Supplementary Data 17). Within the DC3 lineage, we identified four additional clusters, namely the CD14 DC3 (*C1QA, C1QC, CTSB*), CD163 DC3 (*CD163, GIMAP4, TXNIP*), ISG-DC3 (*IFIT3, OAS1, FCGR3A, FRMD3*), and CCR7 DC3 (*LAMP3, TMEM176A/B, PLA2G2D, NR1H3, ME1)* (Fig. 5D, F). Interestingly, the cDC2B lineage in human tumors showed a larger heterogeneity compared to mouse tumor cDC2B, comprising seven clusters, including LDHA cDC2B (*LDHA, RGCC, VEGFA, ALDOA*) which expressed some hypoxia-associated genes, CD1C cDC2B (*CD1C, CCL17*), RUNX3 cDC2B (*RUNX3, EEF1G, EIF5AL1*), IL1B cDC2B (*IL1B, PPP1R15A, DUSP2*), CREM cDC2B (*CREM, AREG, SRGN*), CD1E cDC2B (*CD1E, CLEC10A, FCER1A*), and CX3CR1 cDC2B (*CX3CR1, C3, BHLHE41*) (Fig. 5D, F, Supplementary Data 17). Moreover, four cell populations, not identified in mouse tumors, were found in the human TADC atlas, namely the CD207 cDC2 (also described as Langerhans-like cDC2)[32] (*CD207, FCGBP, CD1A*), RORC DC[88,89] (*RORC, ACY3, PRMD16, TMIGD2*), heat-shock protein (HSP)-cDC2[57] (*HSPA1A, HSPA1B, BAG3*), and CXCL10 cDC2 (*CXCL10, FAM26F, PSME2*), which partially share the ISG signature (Fig. 5D, F). CD207 DCs have been reported to accumulate in various human tumors and peripheral tissues, representing a state that can be acquired by cDC2s and DC3s under the influence of factors such as TGFβ and retinoic acid[4,57]. Moreover, CD207 DCs have been shown to negatively correlate with tumor CD8$^+$ resident memory T cells, T-cell clonality, and the survival of patients treated with immune checkpoint inhibitors[57]. Finally, we identified proliferating cDC2s (Prolif-cDC2: *MKI67, STMN1, TOP2A*), CCR7 cDC2 (*CCR7, LAMP3, TNFRSF18/4, INHBA, SLC7A11*), CCR7 cDC1 (*CCR7, LAMP3, CLEC9A, BTLA, IL12B*), CCR7 cDC2A (*IL4I1, CST7, TNFSF18, MMP12*), and a cluster of CCR7 DCs that could not be attributed to any DC lineage probably due to the masking of the lineage signature by the state signature (Fig. 5D, F, Supplementary Fig. 11H). Annotation of the human TADC subsets/states was validated using gene signatures from presiding studies describing human TADC subsets (Fig. 5E, Supplementary Fig. 11I, Supplementary Data 18). We used the DC5 (AXL$^+$SIGLEC6$^+$ also known as AS DC) score based on blood scRNA-seq data of Villani et al.[24] to identify pDC-like cells, and the CLEC10A$^-$ cDC2A score based on Brown et al.[28] to determine the cDC2A identity within the human TADC atlas (Fig. 5E, Supplementary Fig. 11I). Moreover, both the human and mouse cDC2A clusters had high RUNX3 predicted regulon activity (Supplementary Fig. 12E, F). Interestingly, a high cDC2A score was also observed in CD207 cDC2 and RORC DC, highlighting the gene signature similarity between these populations, and clarifying their close localization to each other in the UMAP (Fig. 5D, E, Supplementary Fig. 11I). Importantly, while CD207 cDC2 expressed several cDC2 scores, RORC DCs expressed only the CLEC10A$^-$ cDC2A score from Brown et al.[28]. As such, the CD207 cDC2 were classified as cDC2. Although based on the gene signature scores RORC DCs would be classified as cDC2A, they lacked the core cDC2 signature including *FCER1A, CD1C*, and *PKIB*. Moreover, recent studies have demonstrated evidence of their lymphoid origin[101]. As cDC2A have a myeloid lineage origin[29], to be congruent with literature we assigned the RORC DC to the general DC category without lineage specification (Fig. 5E). To intricately distinguish between the cDC2B and DC3 lineages, we first assessed the CLEC10A$^+$ cDC2B score based on Brown et al.[28], which, similar to the mouse TADC atlas, was found to be high across the various human cDC2B and DC3 clusters. However, the comparative 'DC2 *versus* DC3' signature scores, generated from both the Villani et al.[24] and Dutertre et al.[30] datasets, showed low values in the DC3 clusters. Moreover, the DC2 scores from Villani et al.[24] and Dutertre et al.[30] were higher in the cDC2A and cDC2B clusters compared to the DC3 clusters. Finally, the comparative 'DC3 *versus* DC2' score as well as DC3 score based on the Dutertre et al.

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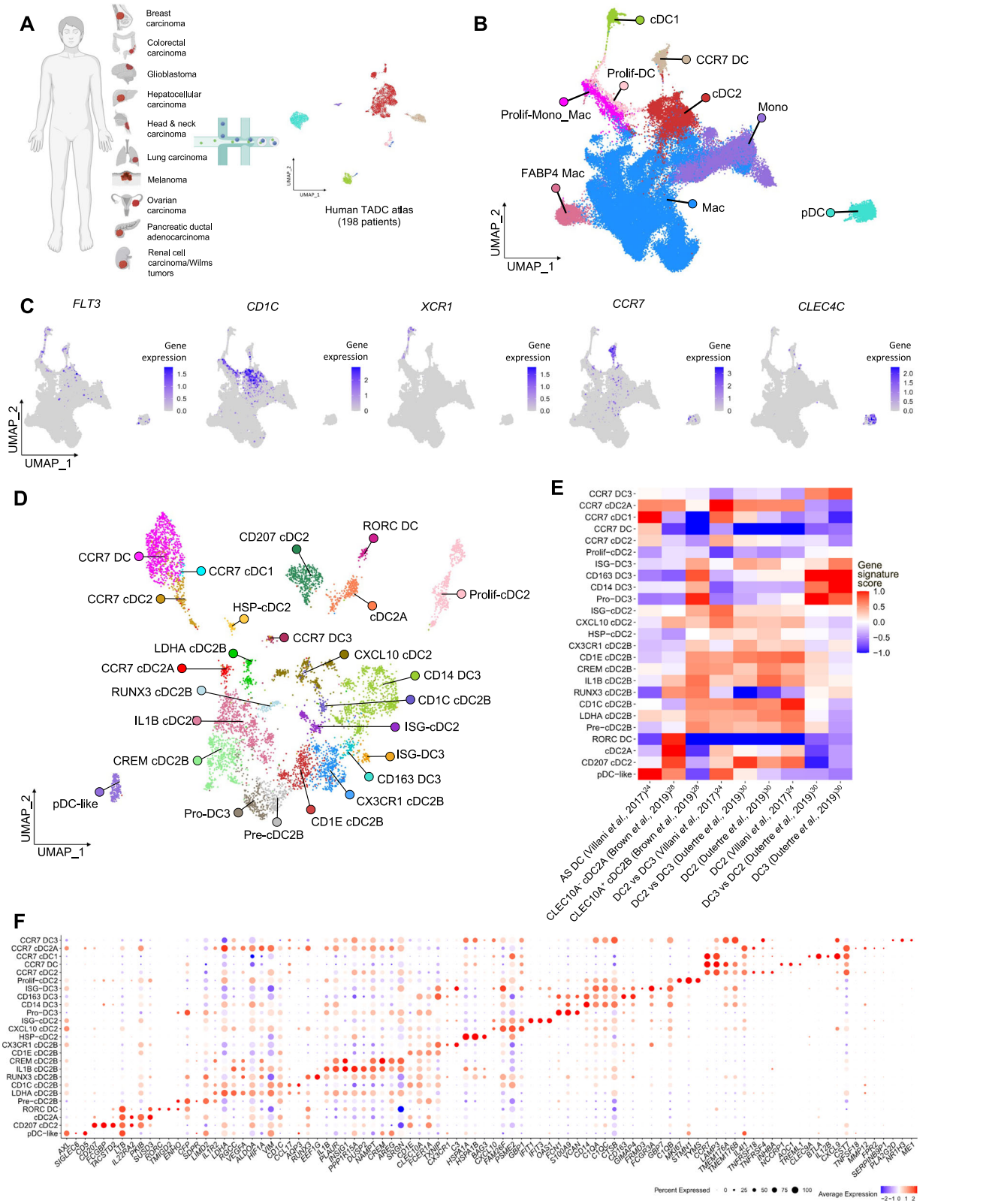


Fig. 5 | Single-cell RNA sequencing reveals 25 DC subsets and states across 10 human cancer types. A Schematic overview of the scRNA-seq datasets used to generate the human TADC atlas. Created in BioRender. Laoui, D. (2026) https://BioRender.com/5vbskux. BC $n = 26$, CRC $n = 17$, GBM $n = 13$, HCC $n = 14$, HNSCC $n = 26$, LC $n = 55$, Melanoma $n = 13$, OC $n = 12$, PDAC $n = 15$, RCC $n = 7$. **B** UMAP plot of 61,432 MNP cells, colored by major cell type. **C** UMAP plots highlighting DC-specific gene expression within the MNP atlas from (**B**). **D** UMAP plot of 6,719 cDC2/DC3/CCR7 DCs, clustered into 25 subsets and states. **E** Heatmap showing gene signature scores in the human TADC atlas, generated using the scRNA-seq datasets of Villani et al.[24], Brown et al.[28], and Dutertre et al.[30], used to annotate the different human TADC subsets and states. **F** Dotplot showing the average expression of DE genes within the different cDC2/DC3/CCR7 DC clusters from (**D**).

blood scRNA-seq data[30] were highest in the DC3 clusters (Fig. 5E, Supplementary Fig. 11I). Notably, the MAF and the MAFB regulons had high predicted regulon activity in both the human and mouse DC3s (Supplementary Fig. 12E, F). Taken together, the high cDC2B, DC2, and 'DC2 *versus* DC3' scores, in combination with the low 'DC3 *versus* DC2' and DC3 only scores enabled us to annotate cDC2B within the human TADC atlas (Fig. 5E, Supplementary Fig. 11I).

When comparing the relative abundance of the respective TADC clusters between mouse and human, we observed higher proportions of CCR7 DCs and DC precursors in mouse, while pDCs, Prolif-cDC2s, cDC2B, and DC3 were enriched in human (Supplementary Fig 11J). This may facilitate the dissection of the greater cDC2B and DC3 heterogeneity observed in humans, while their lower abundance in mice may limit this resolution (Fig. 5D). RORC DC, CD207 cDC2, and HSP-DC were exclusively detected in human, while Cd83 DC and Mgl2 DC were restricted to mice. Further investigation is required to determine whether these clusters represent true species-specific heterogeneity or reflect technical factors, such as differences in total DC abundance due to patient-specific parameters, tissue processing methods, or other experimental variables (Figs. 1F, 5D).

In conclusion, we have generated a human TADC atlas using 10 different cancer types, wherein we fine-map the TADC heterogeneity to reveal 25 tumor-associated cDC2/DC3 subsets and states.

Mouse and human tumors harbor analogous TADC subsets and states

While several studies so far have assessed the correlation of mouse and human DC subsets[38], none have done so encompassing the level of granularity of DC subsets and states of our TADC atlases. We calculated Pearson's correlation coefficients between the mouse and human TADC subsets by comparing the average orthologous gene expression within each species. While some clusters, such as cDC1, pDC, Ccr7 DCs, DC3, and Pro-DC3, showed strong cross-species correlations, others displayed only moderate similarity, indicating partial conservation alongside species-specific differences in TADC gene expression (Fig. 6A, Supplementary Fig. 12A). Our results are supported by previous findings that showed the pDC lineage to be more conserved than the cDCs between mouse and human[102]. Furthermore, a notable correlation was also observed between mouse and human TADCs at the average predicted regulon activity levels (Fig. 6B, Supplementary Fig. 12B). Finally, we determine several orthologous genes that are shared across the different TADC subsets among the two species (Fig. 6C). Overall, we demonstrate that the correlation between mouse and human TADCs was moderate at both the gene expression and predicted regulon activity levels, and the shared similarity patterns may help guide translational insights from pre-clinical models to humans.

Human TADC subsets and states can be associated with patient outcomes

When assessing how the abundance of TADC subsets and states varied across the different cancer types, we found that cDC1 and pDC were most abundant in HCC and HNSCC, respectively (Fig. 7A, Supplementary Fig. 12C, Supplementary Fig. 13A, B). Consistent with mouse cDC2A, human cDC2A were highly abundant in GBM (Fig. 7A, Supplementary fig. 12C, Supplementary Fig. 13A, B). The combined cDC2B compartment was most abundant in RCC, while the CD14 and CD163 DC3 together were present in the highest percentage in CRC and OC (Fig. 7A, Supplementary fig. 12C, Supplementary Fig. 13A, B). Moreover, ISG-cDC2 were predominant in GBM, RCC, and PDAC, whereas ISG-DC3 were most abundant in RCC and OC (Fig. 7A, Supplementary Fig. 12C, Supplementary Fig. 13A, B). HSP-cDC2 and CXCL10 cDC2 were most prominent in HCC and HNSCC, respectively (Fig. 7A, Supplementary Fig. 12C, Supplementary Fig. 13A, B). Furthermore, TADC subsets/states also varied with cancer stage, patient sex, and sample site of tumor collection (Fig. 7B, Supplementary Fig. 13C–E). Notably, in these analyses, we only included the cancer types for which the datasets had sufficient patient numbers for the assessed criteria (Fig. 7B, Supplementary Fig. 13C–E, Supplementary Data 14). Interestingly, while the pan-cancer cDC1 abundance remained constant across different stages, we observed an increase in some cDC2B states (CD1E cDC2B and CX3CR1 cDC2B) in stage IV tumors compared to stages I-III (Fig. 7B). Upon assessing the effect of cancer stage on TADC abundance in LC patients, a sharp increase in cDC2A was observed in stage III (Figure Supplementary Fig. 13C). In CRC patients, a decrease in cDC2A, IL1B cDC2B, and CD14 DC3 was seen in stage III compared to stage II patients, while pDC, pDC-like, CD1E cDC2B, CX3CR1 cDC2B, HSP-cDC2, ISG-DC2, Prolif-cDC1, and CCR7 DC increased with tumor stage (Supplementary Fig. 14C).

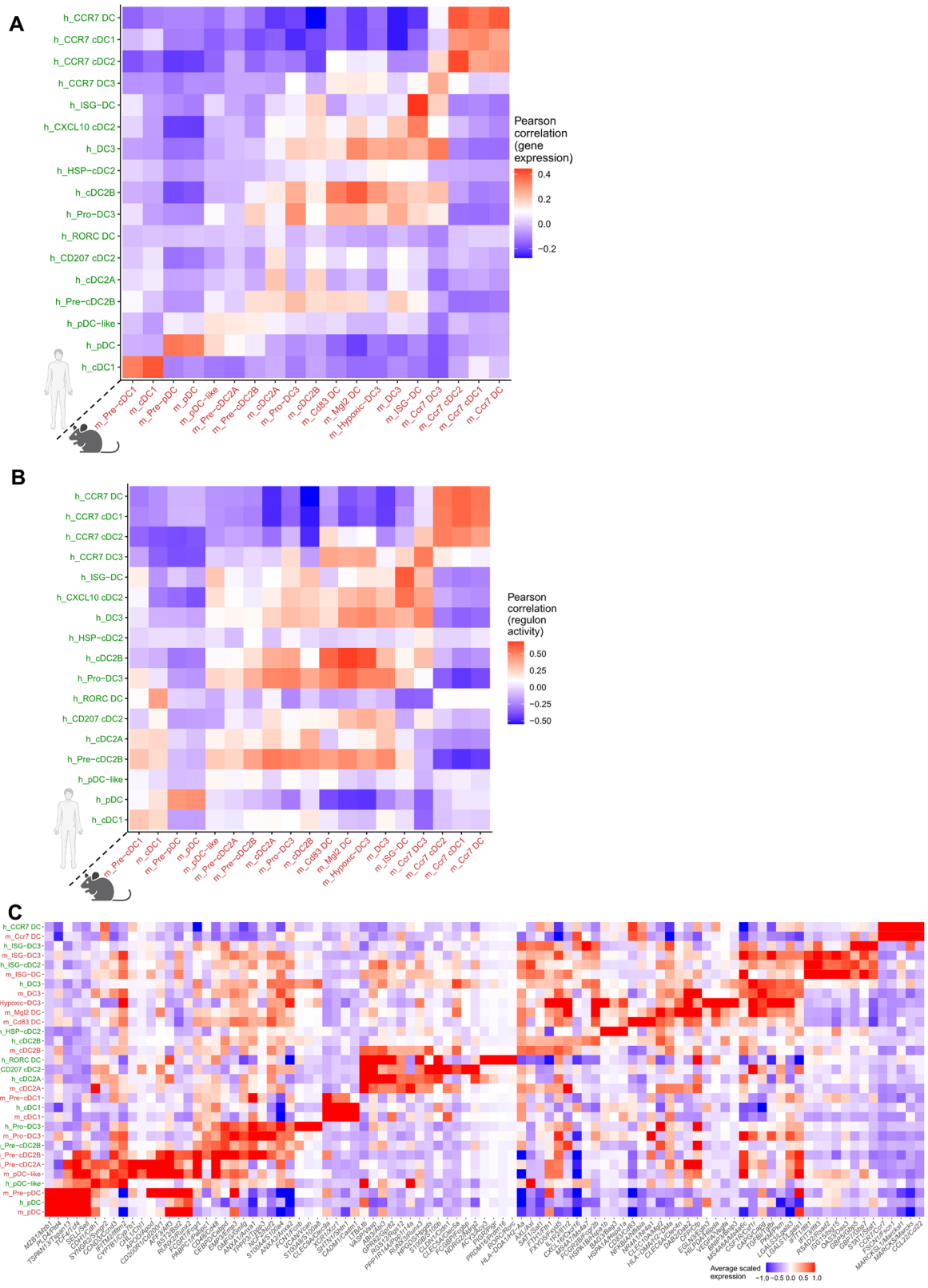
To evaluate whether the kinetic changes observed in the TADC compartment of mouse models are mirrored in human tumors, we analyzed the average expression of ISG and maturation gene signatures across pan-cancer human TADCs broadly categorized as cDC1, cDC2_DC3, pDC, and CCR7 DC (Fig. 7C, D). Consistent with our observations in mouse tumor models, we detected a notable increase in the ISG signature within cDC2_DC3 at stages II and III compared to stage I; however, this was followed by a drop at stage IV (Figs. 7C, 3H, Supplementary Fig. 8A–E). Furthermore, the maturation signature showed an initial increase within tumor cDC2_DC3 followed by a progressive decreased from stages II to IV, in line with our observations in mouse models (Figs. 7D, 3I, Supplementary Fig. 8F–J).

Given the changes in TADC compartments across cancer stages, we assessed whether the different TADC subsets/states can be associated with certain patient outcomes. Hereto, we applied SCIPAC[103], an algorithm that infers a quantitative estimation of cell-phenotype association by matching a single-cell dataset and a bulk dataset with the phenotype of interest. We performed this analysis in all our human TADC clusters in combination with the progression-free survival of 13 large TCGA datasets. This enabled us to assess the association strength of each TADC cluster with the hazard ratio (HR) across the different TCGA bulk datasets (Fig. 7E). Additionally, to identify potential pan-cancer effects, we performed a meta-analysis integrating all 13 different TCGA datasets. Evidently, the prognostic impact of specific DC subsets varied across cancer types (Fig. 7E–H). The meta-analysis revealed that LDHA cDC2B, and CD14 DC3 expression associate with worse prognosis (significantly high association strength with the HR), whereas CCR7 DC predict a better prognosis (significantly low association strength with HR) (Fig. 7E–H). Although we observe associations between TADC and patient outcomes, some discrepancies remain in the results, and some specificity may have been lost due to the limitation that the single-cell and bulk data might not be identically composed. Nevertheless, this approach represents an indicative alignment between our single-cell clustering and patient outcomes from bulk datasets. Collectively, we demonstrate that the abundances of the TADC subset/state vary across the cancer type, stage, and other patient characteristics. More importantly, different TADC subsets/states can be associated with patient outcome.

Discussion

In this study, we have generated pan-cancer scRNA-seq mouse and human atlases, revealing the detailed TADC heterogeneity and fine-mapping DC lineage-defined subsets and functional states. The primary goal of this study was to offer a comprehensive framework that integrates all DC subsets in a unified platform within the cancer context, applicable to both murine and human systems. By doing so, we aimed to provide a powerful tool for researchers to better understand the role of DCs in cancer, fostering greater consistency and facilitating future studies in this critical area. Previous pan-cancer atlases focused predominantly on broad immune compartments in human cancers and provided only limited high-resolution characterization of

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TADCs[32,38,55–57,99,104–107]. The study by Courau et al. represents one of the most extensive cross-species transcriptomic tumor infiltrating immune cell comparisons to date and demonstrated substantial divergence between mouse models and human tumors in immune composition and chemokine networks[108]. However, although comprehensive pan-cancer analyses across multiple mouse models and integrative comparisons between mouse and human have been reported, analyses specifically centered on fine-grained TADC subset/states are currently limited[38,39,109,110]. Our mouse and human TADC atlases comprise of data from 14 cancer models and 10 cancer types, respectively. Both atlases constitute the cDC1, cDC2, DC3, and pDC lineages, and their diverse associated states. Both the cDC2 (including cDC2A and cDC2B) and DC3 compartments were observed to exist also in ISG- and CCR7 DC states, consistent with previous reports for

Fig. 6 | Analogous TADC subsets and states can be found across mouse and human tumors. Heatmaps showing Pearson correlation between the corresponding TADC atlas clusters from mouse and human tumors at **A** gene expression level; **B** predicted regulon activity level, based on gene regulatory network inference using SCENIC. Mouse and human icons created in BioRender. Laoui, D. (2026) https://BioRender.com/jdazeb9. Different subclusters were merged for ease of visualization including h_cDC2B (LDHA cDC2B, CD1C cDC2B, RUNX3 cDC2B, IL1B cDC2B, CREM cDC2B, CD1E cDC2B, CX3CR1 cDC2B), h_DC3 (CD14 DC3, CD163 DC3), h_ISG-DC (ISG-cDC2, ISG-DC3), h_CCR7 cDC2 (CCR7 cDC2, CCR7 cDC2A), m_cDC1 (cDC1, Ccr7 Cxcl9 cDC1), m_ISG DC (ISG-Pro-DC3, ISG-DC3 (1), ISG-DC3 (2), ISG-DC, ISG-cDC2, Ccr7 ISG-DC), m_Ccr7 cDC2 (Ccr7 Pre-cDC2A, Ccr7 cDC2, Ccr7 cDC2A). **C** Heatmap showing average scaled expression of shared mouse and human orthologous genes across the different subsets and states from the mouse and human TADC atlases. Following subclusters were merged: h_cDC2B (LDHA cDC2B, CD1C cDC2B, RUNX3 cDC2B, IL1B cDC2B, CREM cDC2B, CD1E cDC2B, CX3CR1 cDC2B), h_DC3 (CD14 DC3, CD163 DC3), h_ISG-DC (ISG-cDC2, ISG-DC3, CXCL10-cDC2), h_CCR7 DC (CCR7 DC3, CCR7 DC, CCR7 cDC2, CCR7 cDC2A, CCR7 cDC1), m_ISG DC3 (ISG-Pro-DC3, ISG-DC3 (1), ISG-DC3 (2)), m_cDC1 (cDC1, Ccr7 Cxcl9 cDC1), m_ISG DC (ISG-DC, ISG-cDC2, Ccr7 ISG-DC), m_Ccr7 DC (Ccr7 Pre-cDC2A, Ccr7 cDC2, Ccr7 cDC2A, Ccr7 cDC1, Ccr7 DC, Ccr7 DC3). Red, Mouse TADC clusters; green, human TADC clusters.

cDC1 and cDC2[32,33,42,52,57,111]. Validation of cluster annotations was based on gene signature scores obtained from fate-mapping studies on DC ontogeny, thereby efficiently and in an unbiased manner characterizing the DC populations based on entire gene signatures in addition to a selection of marker genes. Interestingly, DC progenitor states, including Pre-cDC1, Pre-pDC, pDC-like cells, Pre-cDC2A, Pre-cDC2B, and Pro-DC3, were identified within the TME in both mouse and human. As many of these progenitors partly share transcriptional profiles with their corresponding mature DCs populations, they are often overlooked in low-resolution analyses, underscoring the utility of the gene signatures identified in this study.

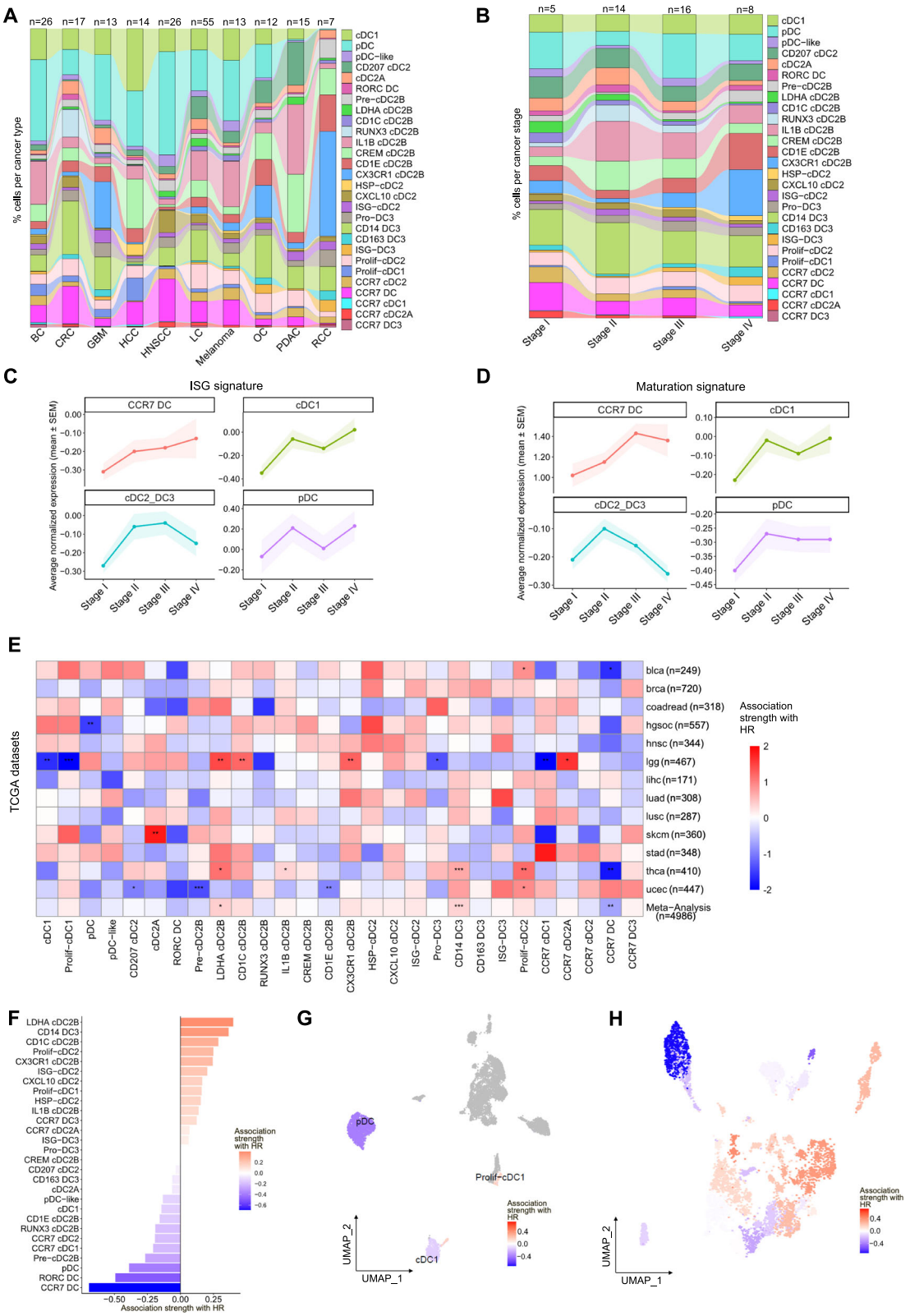
Our mouse atlas constituted two TADC clusters, namely the Cd83 DC and Mgl2 DC, that showed the expression of both cDC2 and DC3 gene signature scores and hence could not be attributed to either DC lineage. These clusters were not identified in the human TADC atlas. Contrastingly, our human TADC atlas revealed a broad diversity of cDC2 (particularly cDC2B), and DC3 states, along with CD207 cDC2, CXCL10 cDC2, RORC DC, and HSP-cDC2, all of which were not observed in the mouse atlas. Interestingly, the mouse atlas showed a higher proportion of DC precursors and CCR7 DCs, but a lower proportion of pDCs, Prolif-cDC2s, cDC2B, and DC3. These differences likely facilitate the dissection of the underlying cellular diversity resulting in a higher heterogeneity in compartments enriched in each species, such as cDC2B and DC3 in humans. Importantly, variations in the total DC abundance may be supplemented by technical factors, including differences in tissue collection and processing, patient treatment history, genetic background, cancer types and stages, as well as demographic characteristics, including age, sex, and lifestyle, which were unavailable for most patients[112–114] and can skew the comparison of heterogeneity across species.

When analyzing the influence of cancer stage on TADCs in mouse kinetic EO771, KPCY, B16F10, and GBM datasets, we observed a decrease in DC maturation genes within the cDC2_DC3 compartment in larger tumors, which has also been reported in multiple myeloma[115]. In contrast, KP tumors showed an increase in cDC2_DC3 maturation, potentially because these tumors were processed together with the adjacent lung. Notably, peritumoral lung tissue from NSCLC patients has been shown to harbor more mature DCs than tumor tissue[116], suggesting that the increase in maturation may reflect contributions from adjacent lung tissue. That said, it is important to note that this is a limitation that extends to several mouse tumor models currently used, where tumor nodules are not always fully dissectible from the surrounding tissue. Future studies could overcome this limitation by applying spatial analysis to tumor-bearing organs. PanCK and DAPI staining would enable clear delineation of tumor regions based on the higher cellular density compared to adjacent tissues. Subsequent fluorescence immunohistochemistry using markers such as CD11c, MHC-II, CCR7, CD40, CD80, Ly6C, and F4/80 could be used to assess DC activation states while excluding macrophages/monocytes within defined spatial compartments. In addition, spatial transcriptomics would allow a higher correlation with single-cell datasets by enabling direct comparison of gene expression patterns.

In the mouse kinetic datasets, we observed that the ISG signature increased in the cDC2/DC3 compartment with tumor progression, which was also observed in stages II and III human cancers. Type-I interferons are known to regulate DC turnover, mediate DC activation, prolong DC survival, aid in their retention of tumor antigens, and facilitate $CD4^+$ priming and $CD8^+$ T-cell cross-priming[42,52,117–121]. Contrastingly, although interferons are known to promote DC maturation, chronic interferon signaling can lead to DC dysfunction, as seen in persistent viral infections[122,123]. This may be in line with our findings of increased ISG and reduced DC maturation signatures in the cDC2_DC3 compartment during tumor progression. In human tumors, we observed an initial increase in ISG-signature from stages I to III, followed by a decline at stage IV. Similarly, monocytes from chronic HIV-1-infected individuals showed an eventual loss of interferon response, which correlated with lower peripheral blood $CD4^+$ T-cell counts and more advanced disease stage, suggesting a potential mechanism of immune evasion through desensitization to interferon signaling[124].

Interestingly, in EO771, KP, and GBM tumors, the Cd1d1 Ccr7 DCs were enriched in early versus late tumor stages. In B16F10 tumors, although this cluster was not identified (likely due to limited cell numbers in the dataset), *Cd1d1* gene was expressed highest in Ccr7 DCs in early-stage tumors. Mouse CD1d1 (homolog of CD1d in human) is a non-classical MHC-I-like protein involved in (glyco)lipid antigen presentation to T cells. Moreover, CD1d-expressing DCs are critical in stimulating iNKT cells[125–127], known to be immune regulatory in autoimmune disorders like diabetes[128–130]. Contrastingly, CD1d deficiency in LLC tumor-bearing mice promoted tumor growth, and lung cancer patients with more intra-tumoral $CD1d^+$ DCs had more activated T and NKT cells[131]. Interestingly, a TCGA analysis revealed that *CD1D* expression decreased with tumor progression[131], and accordingly, circulating DCs from advanced-stage melanoma and RCC patients showed poor activation of iNKT cells[132]. This is in concurrence with our findings, where we predominantly detected Cd1d1 Ccr7 DCs in early-stage tumors, indicating their potential role in inducing early anti-tumor responses. In contrast, we found that Cd1d1 Ccr7 DCs in KPCY tumors were predominantly present in late-stage tumors. TCGA analysis of PDAC showed that *CD1D* expression is associated with improved overall survival[131]. However, in a mouse model of colitis, CD1d-mediated activation of NKT cells led to the secretion of chemokines such as CXCR2 ligands (CXCL1, 2, 3) by the colon epithelium[133]. CXCR2-dependent neutrophil recruitment has been reported in both human PDAC and mouse KPC/KPCY tumors, where higher levels of tumor-associated neutrophils correlate with poor prognosis[134,135]. As such, our observation of increased Cd1d1 Ccr7 DCs in late stage KPCY tumors may reflect such indirect pro-tumoral effect leading to neutrophil accumulation in the tumor. Overall, while *Cd1d1* expression was predominant in early stages of cancer in four out of five mouse models, model-specific variations warrant further investigation. In our kinetic EO771 analysis, we found that Hif1a Mgl2 DCs were only enriched in early EO771 tumors, which may also reflect a model-specific feature such as an IFN-driven or lipid-mediated metabolic reprogramming[121,136], or a response to the orthotopic inoculation of rapidly growing cancer cells.

https://doi.org/10.1038/s41467-026-73721-w



Tumor-driven extramedullary hematopoiesis can occur in multiple organs, including the spleen and LNs[137–142]. Reprogramming of DC development within the BM is seen in mouse models and patients with BC and PDAC[143]. In accordance, we found tumor-reprogrammed (TR) DC3, ISG-cDC2, Cd83 DC, and Hypoxic-DC clusters that were predominantly enriched within the tumors and LN from tumor-bearing mice. Although the function of these TR DC subsets and states is not fully characterized, they consistently showed an increase in ISGs and inflammation-associated genes compared to their non-TR counterparts.

Multiple studies have investigated the relationship between tumors immune composition and patient prognosis. Across cancer types, immune-rich tumors generally associate with improved survival, although it was shown that distinct immune archetypes confer

Fig. 7 | Human TADCs subsets and states vary across cancer type and cancer stage, and can be associated with patient outcome. A, **B** Alluvial plot showing the percentage of human TADC atlas clusters across **A** different cancer types; **B** different cancer stages. **C**, **D** Ribbon plots highlighting changes of the expression within TADC subsets and stages of **C** selected interferon-stimulated and inflammation-associated genes (refer to "Methods"); **D** selected DC-activation and maturation genes (refer to "Methods"). The solid line represents the average expression across genes, and the shaded ribbon indicates the standard error (SEM) calculated across genes. **E** Heatmap representing the association strengths of the different DC subsets and states with the hazard ratio (HR) in 13 TCGA bulk datasets: Bladder Urothelial Carcinoma (blca), Breast Invasive Carcinoma (brca), Colorectal Adenocarcinoma and rectal adenocarcinoma (coadread), Head and Neck Squamous Cell Carcinoma (hnsc), Brain Lower Grade Glioma (lgg), Lung Adenocarcinoma (luad), Lung Squamous Cell Carcinoma (lusc), Stomach Adenocarcinoma (stad), Thyroid Carcinoma (thca), Uterine Corpus Endometrial Carcinoma (ucec), Liver Hepatocellular Carcinoma (lihc), High-Grade Serous Ovarian Cancer (hgsoc), Skin Cutaneous Melanoma (skcm), and the meta-analysis of all 13 datasets. Red or blue represents a positive (worse predicted survival), or negative association with HR (good predicted survival), respectively, and the opacity represents the association strength. *p*-values are represented on the plot as *, **, *** for *p*-value < 0.1, <0.05 and <0.01 respectively. Two-sided *p*-values were calculated by the SCIPAC nonparametric bootstrap strategy for the individual TCGA datasets. No correction for multiple testing was performed. Source data are provided as a Source Data file. **F** Barplot of the association strength with HR per DC subtype in the meta-analysis performed in (**E**). Red or blue represents a positive or negative association with HR, respectively, and the opacity represents the association strength. A strong positive association (red) indicates a positive association with HR and a worse survival; **G**, **H** UMAP showing **G** all the DCs from the human TADC atlas from Supplementary Fig. 11B; **H** subsetted DC2/DC3/CCR7 DCs from Fig. 5D color-coded for their association strength with the HR based on the meta-analysis on 13 TCGA datasets. Red or blue represents a positive (worse survival) or negative (better survival) association with HR, respectively, and the opacity represents the association strength.

prognostic benefit depending on the cancer type[144]. In luminal breast cancer (LBC), pDCs, cDC1s, and cDC2s have been associated with favorable prognosis, while monocytes and macrophages correlate with poor prognosis in both LBC and triple negative breast cancer (TNBC)[145]. In TNBC, cDC1s associate with good prognosis, while cDC2 and pDCs show limited prognostic value[145]. Our study extends these findings by directly mapping single-cell DC subclusters to bulk tumor transcriptomes and survival data from TCGA, thereby providing a higher-resolution understanding of how specific DC subsets and states relate to patient outcomes. Importantly, rather than relying on gene signatures to correlate TADCs with patient outcome, we adopted a data-driven approach as we found that selecting pre-defined up- and down-regulated genes often resulted in signatures that overlapped with other, non-specific cell types. In addition, such approaches introduce user bias through the arbitrary selection of thresholds for defining "top" differentially expressed genes. To overcome these limitations, we utilized the SCIPAC package, which generates a shared latent space based on the entire transcriptional signature/pattern of single cells, rather than a limited gene subset[103]. This model learns dominant patterns from the single-cell data and then projects bulk RNA-seq samples into the same latent space. This allows identifying which single-cell populations are most closely associated with phenotypes obtained from bulk datasets[103], in our case, the hazard ratios. When analyzing the potential correlation between different TADC subsets/states with patient prognosis, we found that LDHA cDC2 and CD14 DC3 were significantly associated with worse patient outcome, while CCR7 DC were associated with a significantly good patient outcome. The association between CCR7 DC and favorable prognosis has been well established[146–149]. In contrast, a CD1c$^+$CD14$^+$ cell population with features of both DCs and monocytes has been reported to exert immunosuppressive effects in melanoma patients by suppressing CD4$^+$ T cells, and potentially attenuating the efficacy of DC vaccines[150]. Similarly, DC3-like CD1c$^+$CD14$^+$ cells that were derived from cDC2s, were found to be elevated in melanoma patients and to correlate with poor survival among those receiving DC vaccinations[151]. Supporting these observations, studies using a 3D DC-tumor co-culture system demonstrated that the tumor-derived cDC2s can acquire a DC3-like phenotype and were characterized by defective maturation marker expression and impaired Tcell activation capacity[152]. Although some discrepancies remain in the results of our analysis, it represents the closest alignment currently achievable between our single-cell clustering and bulk datasets with patient outcomes.

Our mouse and human TADC subsets and states showed a moderate correlation at the gene expression level, as has been previously reported for several immune cell types, including DCs[39,102,153–155]. Overall, owing to their pivotal roles in driving anti-cancer immunity, DCs are promising candidates for immunotherapy, including in vivo DC targeting and adoptive transfer of DCs (DC vaccination). Due to the overall poor T-cell stimulation capacity and patient benefit of monocyte-derived cell (commonly referred to as MoDC) based vaccines, there is a growing interest in cDC-targeted strategies, owing to their higher MHC molecule expression and T-cell stimulation potential[46,156–161]. A thorough understanding of the TADCs in human cancers and the mouse models used to study them could therefore be an invaluable resource to better design these therapies.

Methods

Mouse strains

All mouse experiments were conducted in accordance with ethical standards and approved by the relevant regulatory bodies at each involved institution. The maximal tumor size/burden permitted by the regulatory bodies was not exceeded, and all humane endpoint criteria were complied with.

Datasets generated in the Laoui lab at Vrije Universiteit Brussel, Belgium: For LLC-OVA, EO771, and B16F10, 6- to 8-week-old female C57BL/6JRj mice were purchased from Janvier Labs (France). The experimental procedures, follow-up, and housing complied with the Belgian Council for Laboratory Animal Science guidelines. Mice were housed at 20–22 °C, 45–65% humidity with 12 h light/dark cycle. The maximal permitted tumor volume of 2000 mm^3 was not exceeded for any of the models. All animal experiments were approved by the Ethical Committee for Animal Experiments of the Vrije Universiteit Brussel (16-220-02, 20-220-26, 20-220-05, 23-220-47).

Datasets generated in the De Palma lab at EPFL, Switzerland: The *Kras*$^{LSL\text{-}G12D/+}$; *Trp53*$^{fl/fl}$ (KP) transgenic mouse model of NSCLC was established by DuPage et al.[162] and bred in-house. KP mice were then crossed with *Msh2*$^{fl/fl}$ mice to reproduce *Kras*$^{LSL\text{-}G12D/+}$; *Trp53*$^{fl/fl}$; *Msh2*$^{fl/fl}$ (KPM) mice in a C57BL/6 background, as previously described[163]. All mice were monitored and housed in accordance with Swiss regulations for the care and use of mice in experimental research (20–24 °C, with 45–65% humidity, and on a 12 h light/dark cycle). 13 to 14-week-old mice (both genders) were used to induce lung carcinoma. The termination criteria included (i) individual lung tumors exceeding a diameter of 3 mm, as assessed by micro-CT imaging; (ii) diffuse lung inflammation; and (iii) dyspnea. None of the mice met these criteria. All procedures on the KPM mice were conducted according to protocols approved by the Veterinary Authorities of the Canton Vaud, in obedience to the Swiss laws (protocol VD3785a-g).

Datasets generated in the Close and Blomme labs at GIGA-University of Liège, Belgium: For the *Kras*$^{LSL\text{-}G12D/+}$; *Trp53*$^{fl/fl}$ (KP) model, 10-week-old mice (both genders) were used for AAV infection and further analyzed after 5 or 8 weeks of tumor development. *Kras*$^{LSL\text{-}G12D/+}$ were purchased from the Jackson Laboratory (strain B6.129S4-Krastm4Tyj/J). *Trp53*$^{fl/fl}$ mice were a kind gift from Prof. C.J. Desmet (strain B6.129P2-Trp53tm1Brn/J, GIGA-University of Liège)[164]. KP mice were generated by breeding *Kras*$^{LSL\text{-}G12D/+}$ with *Trp53*$^{fl/fl}$ mice. Euthanasia

criteria for KP mice were determined based on overall body condition scoring, including fur coat condition, loss of more than 20% body weight, activity (basal and provoked), and fast abdominal breathing. None of these criteria was exceeded. 12-week-old female MMTV-PyMT were used. All experimental procedures, follow-up, and housing complied with the Belgian Council for Laboratory Animal Science guidelines. The maximal permitted tumor volume of 1000 mm^3 was not exceeded for the MMTV-PyMT model. All animal experiments on the KP and MMTV-PyMT mice were approved by the Ethical Committee for Animal Experiments of the University of Liège (ethical protocol numbers 2587 and 2267). All mice were bred and housed in the conventional mouse facility of the University of Liège at 20–24 °C, with relative humidity of 40–60% and maintained on a 12 h light/dark cycle.

Tumor models

The mouse TADC atlas was built using a carefully selected set of in house-generated and publicly available datasets based on quality, interpretability, and compatibility criteria. Inclusion was restricted to datasets with adequate DC representation, reliable metadata, and data generated on the 10X Genomics or BD Rhapsody platforms to minimize batch effects. Many publicly available datasets were excluded because they had been pre-sorted for non-immune cells ($CD45^-$) or specific immune cell populations (e.g., B cells, macrophages), or because they contained unsorted whole-tumor samples, in which DCs, which represent a relatively rare population, were underrepresented, making integration of such datasets unreliable.

scRNA-seq/CITE-seq datasets of LLC, EO771, B16F10, KP_IV, EMT6-luc, 4T1-luc, MMTV-PyMT, KPC, MC38, STOSE, and GL261 used in the mouse TADC atlas were taken from previously published studies[33,35,43,58–66]. The LLC-OVA, KPM, KP (early and late stage), a second B16F10, additional MMTV-PyMT, and the EO771 kinetic datasets were newly generated. Kinetic GBM (CAG-LSL-$EGFR^{WT}$;Ink$\Delta 2/3^{-/-}$; $PTEN^{2lox}$), B16F10, and KPCY datasets were obtained from published studies[82–84].

Day 20 EO771 tumor samples from the kinetic datasets (similar tumor stage as the previously published EO771 dataset[59]) were included within the TADC atlas to increase the number of EO771 TADCs. EO771 cells were kindly provided by Prof. Massimiliano Mazzone (KU Leuven, Belgium) and subsequently cultured in DMEM (Gibco cat# 41965–039) supplemented with 10% (v/v) heat-inactivated fetal calf serum (FCS; Capricorn Scientific, cat# FBS-12A), 300 µg mL^{-1} L-glutamine (Sigma-Aldrich, cat# G8540), 100 units mL^{-1} penicillin and 100 µg mL^{-1} streptomycin (Gibco, cat#15140122). A single-cell suspension of 500.000 EO771 cells resuspended in 50 µL HBSS (Gibco, cat#14175129) was injected orthotopically in the fourth inguinal MFP and allowed to grow. Tumors were harvested on day 9 (~27.75 mm^3), day 14 (~259.6 mm^3), and day 20 (~627.98 mm^3) post-inoculation.

The B16F10 cell line (from ATCC) was cultured in DMEM (Gibco, cat# 41965–039) supplemented with 10% (v/v) heat-inactivated FCS (Capricorn Scientific, cat# FBS-12A), 300 µg mL^{-1} l-glutamine (Sigma-Aldrich, cat# G8540), 100 units mL^{-1} penicillin, and 100 µg mL^{-1} streptomycin (Gibco, cat#15140122). A single-cell suspension of 1 million B16F10 cells resuspended in 200 µL HBSS (Gibco, cat#14175129) was injected subcutaneously into the right flank of syngeneic female C57BL/6 mice. The tumors were allowed to grow for 15 days after which they were harvested and processed.

The LLC-OVA cell line was kindly provided by Prof. Dmitry Gabrilovich (The Wistar Institute, Philadelphia, USA). LLC-OVA cells were maintained in RPMI-1640 medium (Sigma-Aldrich, cat# 52400–025) supplemented with 10% (v/v) heat-inactivated fetal calf serum (FCS; Capricorn Scientific, cat# FBS-12A), 300 µg ml^{-1} L-glutamine (Sigma-Aldrich, cat# G8540), 100 units ml^{-1} penicillin and 100 µg ml^{-1} streptomycin (Gibco, cat#15140122). A single-cell suspension of 1 million LLC-OVA cells in HBSS (Gibco, cat#14175129) was injected subcutaneously into the right flank of mice. The tumors were allowed to grow for 14 days after which they were harvested and processed.

The autochthonous KPM lung carcinoma model was induced in 13–14-week-old male and female mice. Sedation was carried out with intraperitoneal injection of medetomidine hydrochloride (1 mg/kg; Domitor, Orion Pharma) and ketamine (100 mg/kg; Sintetica) in sterile saline (B. Braun Ecotainer, cat#3570110), and ocular gel (Viscotears, Novartis) was used to protect the cornea. Mice were kept on a heating pad after sedation and during the whole following procedure. Intratracheal delivery of lentiviral vectors that carried a Cre recombinase (Cre-LVs) was performed as previously described[165]. In brief, around 2000 transduction units of Cre-LVs were diluted in 80 µL of sterile PBS (Gibco, cat#10010023) and delivered intratracheally with a catheter. After intratracheal delivery, mice were injected subcutaneously with atipamezole hydrochloride (2.5 mg/kg; Antisedan, Orion Pharma), a medetomidine hydrochloride antidote, to recover from sedation. Upon Cre delivery, transduced lung epithelial cells start to express $Kras^{G12D}$ (mutated oncogene) and will lose expression of *Trp53* (tumor suppressor gene) and *Msh2* (DNA mismatch repair gene). These genetic changes initiate lung tumors in each mouse. IgG2a antibodies (10 mg/kg; Bio X Cell, clone 2A3, cat#BP0089) were administered 115, 118, 124 and 127 days post-tumor initiation.

For the spontaneous KP model, KP mice were infected intratracheally with Cre-containing AAV (pAAV6.2ff-mSPB-Cre-IRES-mCherry, 1×10^{12} units per mouse, all vectors produced in-house) to develop lung tumors.

Sample preparation for single-cell RNA-sequencing

LLC-OVA (3 pooled female mice) and EO771 tumors (Day 9: 12 pooled female mice; Day 14: 4 pooled female mice; Day 20: 4 pooled female mice) were excised and collected into 30 µM Actinomycin D (ActD) (Sigma-Aldrich, cat# A1410-2MG) in RPMI, cut into small pieces, incubated with 10 U/mL collagenase I (Worthington, cat# CLSS-1), 400 U/mL collagenase IV (Worthington, cat# CLSS-4), 30 U/mL DNase I (Worthington, cat# DCLS), and 15 µM ActD in RPMI for 20–25 min at 37 °C. The suspension was then squashed with the back of a 10 mL syringe plunger (Greiner, Omnifix, cat# 473203), triturated, and filtered on a 70 µm nylon filter.

B16F10 tumors (10 pooled female mice) and whole tumor-bearing lungs from KP mice (2 males and 2 females pooled for the early stage tumors and 3 males and 1 female pooled for the late stage tumors) were excised and collected into 30 µM Actinomycin D (ActD) (Sigma-Aldrich, cat# A1410-2MG) in RPMI, cut into small pieces, incubated with 10 U/mL collagenase I (Worthington, cat# CLSS-1), 400 U/mL collagenase IV (Worthington, cat# CLSS-4), 30 U/mL DNase I (Worthington, cat# DCLS), and 15 µM ActD in RPMI in a ThermoMixer C (Eppendorf) (1500 rpm, 20 min, 37 °C).

Spleens (3 pooled female naive mice and 3 pooled female LLC-OVA mice) were collected in 30 µM ActD and were squashed between 70 µm nylon filters with the back of a 10 mL syringe plunger (Greiner, Omnifix, cat# 473203). LNs (inguinal tumor-draining LN pooled from 3 female LLC-OVA bearing mice and naïve inguinal LN pooled from 2 female naïve mice) were collected in 30 µM ActD, cut into small pieces, incubated with 10 U/mL collagenase I (Worthington, cat# CLSS-1), 400 U/mL collagenase IV (Worthington, cat# CLSS-4), 30 U/mL DNase I (Worthington, cat# DCLS), and 15 µM ActD in RPMI for 40 min at 37 °C, vortexing every 10 min followed by filtration through a 70 µm nylon filter. After centrifugation at 450 ×*g*, if needed, the single-cell suspensions of all organs were then treated with ammonium–chloride–potassium (ACK) erythrocyte lysis buffer. All subsequent processing was performed in RPMI or HBSS buffers with 3–5 µM ActD.

The single-cell suspensions of desired samples were stained for fluorescence-activated cell sorting (FACS) with the appropriate fluorescence-conjugated antibody cocktails, a mouse cell surface

protein antibody panel containing oligo-conjugated antibodies for CITE-seq and TruStain FcX™ PLUS Antibody (clone S17011E, Biolegend, cat# 156604, 1:30 dilution) in phosphate-buffered saline (PBS; Gibco, cat# 20012027) with 0.04% bovine serum albumin (BSA; Invitrogen, cat# AM2616) for 20 min on ice. After incubation, cells were incubated with 7-Amino-actinomycin D (7-AAD; Biolegend, cat# 420404, 1 μL/10^6 cells) for 5 min before sorting.

LLC-OVA tumors, B16F10 tumors, KP tumors, and LN samples were stained with anti-CD45 APC-Cy7 (clone 30-F11, Biolegend, cat# 103116, 1:666 dilution) and sorted for $CD45^+7\text{-}AAD^-$. LLC-OVA, KP, and LN samples were sorted on the BD FACSAria™ II, while B16F10 samples were sorted using the BD FACS Symphony S6.

EO771 samples were stained with anti-CD45 APC-Cy7 (clone 30-F11, Biolegend, cat# 103116, 1:666 dilution), anti-CD11c PE (clone HL3, BD Biosciences, cat#561044, 1:666 dilution) and anti-MHCII AF488 (clone M5/114.15.2, Biolegend, cat# 107616, 1:666 dilution). $CD45^+7\text{-}AAD^-MHCII^{low\text{-}hi}CD11c^{low\text{-}hi}$ cells were sorted on a BD FACSAria™ II. Due to the high percentage of live immune cells in spleens, these samples were not sorted.

For KPM tumors, the mice were euthanized and perfused by cardiac injection of PBS. Ten lung tumors of various sizes were harvested from 5 mice (3 males and 2 females) and were then digested using the mouse tumor dissociation kit (Miltenyi, cat#130-096-730) and gentleMACS dissociator system (GentleMACS Octo Dissociator with heaters: Miltenyi, cat#130-096-427; GentleMACS C Tube: Miltenyi, cat#130-093-237) according to the manufacturer's procedure. Digested samples were processed through a 70 μm cell strainer (Corning, cat#431751) and cleaned with PBS with 0.04% BSA before removing the red blood cell using RBC lysing buffer Hybri-Max (Sigma-Aldrich, cat#R7757). Subsequently, dead cells were removed by using a dead cell removal kit (Miltenyi, cat#130-090-101) to obtain the final single cell suspension. Finally, lung tumor-derived cells were washed, filtered through a 40 μm Flowmi strainer (Bel-Art, Wayne, NJ), and then resuspended in PBS with 0.04% BSA. These samples were not sorted.

For MMTV-PyMT, tumors were isolated from the mammary gland (2 pooled female mice), collected and washed with PBS and transferred to a freshly prepared digestion buffer, containing HBSS supplemented with 5% FBS, DNAse I (0.1 mg/mL; Roche, cat# 04716728001) and collagenase A (3 mg/mL; Roche, cat# 10103586001). GentleMACS Octo dissociator from Miltenyi was used following the program for tumor digestion: 37C_m_TDK_2. Red blood cell lysis was performed using the Miltenyi buffer (Miltenyi, cat#130-094-183). Cells were filtered using a 70 μm cell strainer and stained using anti-CD45-Pacific blue (clone 30-F11, Miltenyi, cat#130-110-802, 1:50 dilution) and viability dye AmCyan (Miltenyi, cat # 130-110-206, 1 μL/10^7 cells). Live $CD45^+$ cells were sorted using FACS Aria III with nozzle 100 at 3000 events/s.

Single-cell RNA sequencing (scRNA-seq) using the 10x Genomics platform and the BD Rhapsody platform

For tumors from EO771, LLC-OVA, KP, KPM, MMTV-PyMT models, and spleens and LNs from LLC-OVA-bearing and naïve mice, sorted single-cell suspensions were loaded on a Chromium Controller (10x Genomics) to generate single-cell gel beads in emulsion (GEM). Within these samples, CITE-seq was performed on the KP tumors.

For LLC-OVA samples (tumor and tumor-draining LN) and naïve LN, GEMs and scRNA-seq libraries were prepared using a Chromium Next GEM Single Cell 3′ GEM, Library & Gel Bead Kit (v3) and a Chromium i7 Multiplex kit (10x Genomics, PN-120262) according to the manufacturer's instructions.

For late KP tumors (week 8 post Cre-induction) and EO771 tumors, GEMs and scRNA-seq libraries were prepared using a Chromium Next GEM Single Cell 3′ GEM, Library & Gel Bead Kit (v3.1) and a Chromium i7 Multiplex kit (10x Genomics, PN-120262) according to the manufacturer's instructions.

For the early KP tumor, LLC-OVA spleen and naïve spleen samples, GEMs and scRNA-seq libraries were prepared using a Chromium Next GEM Single Cell 5′ GEM, Library & Gel Bead Kit v1.1 and a Chromium Single Cell 5′ Library Construction Kit (10x Genomics, PN-1000020) and a Chromium i7 Multiplex Kit (10x Genomics, PN-120262) according to the manufacturer's instructions.

Briefly, GEM reverse-transcription incubation was performed in a thermal cycler at 53 °C for 45 min, 85 °C for 5 min, and ending at 4 °C. Next, GEMs were broken and complementary DNA (cDNA) was cleaned up using DynaBeads MyOne Silane Beads (ThermoFisher Scientific, cat#37002D) and SPRIselect Reagent kit (Beckman Coulter, cat#B23318). Full-length, barcoded cDNA originating from the mRNA, and from the oligonucleotide-labeled cell surface protein antibodies (for CITE-seq KP samples), was PCR amplified with a 96-deep-well reaction module at 98 °C for 3 min, 14 cycles at 98 °C for 15 s, 67 °C for 20 s and 72 °C for 1 min, 1 cycle at 72 °C for 1 min and ending at 4 °C. For the 5′ samples, the PCR amplification was performed according to the following protocol: 98 °C for 45 s, 14 cycles at 98 °C for 20 s, 67 °C for 30 s and 72 °C for 1 min, 1 cycle at 72 °C for 1 min and ending at 4 °C. Size selection via a SPRIselect Reagent kit was used to separate the amplified cDNA molecules for 3′ gene expression library construction. Gene expression library construction to generate Illumina-ready sequencing libraries was performed after clean-up using a SPRIselect Reagent kit and enzymatic fragmentation by adding R1 (read 1 primer), P5, P7, i7 sample index and R2 (read 2 primer sequence) via end-repair, A-tailing, adapter ligation, post-ligation SPRIselect clean-up/size selection and sample index PCR. The cDNA content of pre-fragmentation and post-sample index PCR samples was analyzed using a 2100 BioAnalyzer (Agilent). Sequencing libraries were loaded on an Illumina NovaSeq 6000 flow cell, with sequencing settings according to the recommendations of 10x Genomics.

For the B16F10 tumors, sorted single-cell suspensions were loaded on the BD Rhapsody 8-lane cartridge using BD enhanced cartridge reagents v2, cDNA and WTA amplification kit (BD Biosciences). Reverse transcription, cDNA amplification and library construction were performed following the manufacturer's instructions (23-24117(02)). Libraries were sequenced on Illumina NovaSeq 6000 at the VIB Nucleomics Core (Leuven).

For KPM tumors, cells were checked for the absence of significant doublets or aggregates and loaded into a Chromium X Single Cell Controller (10x Genomics) in a chip together with beads, master mix reagents (containing RT enzyme and poly-dt RT primers) and oil to generate single-cell-containing droplets. For each sample, 20000 cells were targeted. Single-cell gene expression libraries were then prepared using the Chromium Single Cell 3′ Library & Gel Bead Kit v4 (PN-1000691) following the manufacturer's instructions (protocol CG000731 Rev A). Quality control was performed with a TapeStation 4200 (Agilent) and QuBit dsDNA high sensitivity assay (Thermo) following manufacturer instructions. The sequencing libraries were loaded on a NovaSeq 6000 (Illumina) flow cell and sequenced according to manufacturer instructions, with a read1 of 28nt and a read2 of 90nt, at a depth of about 50-100 thousand reads/cell.

For PyMT tumors, library construction was with 10x Genomics platform Chromium™ Single Cell 3′ Library & Gel Bead Kit v2, 16 rxns PN-120237. Cellular suspensions were loaded on the Chromium Controller (10x Genomics, Pleasanton) to generate Gel Bead-In-Emulsions (GEMs). Barcoded sequencing libraries were generated with the Chromium Single Cell 3′ Reagent Kits v2 (10x Genomics) following manufacturer's instructions. Sequencing libraries were loaded at 1.4 pM on an Illumina NextSeq500 with NextSeq 500/550 Mid Output v2 kit (150 cycles; Illumina) with the following read lengths: 26 base pairs (bp) for Read 1 (16-bpBarcode + 10-bp Randomer), 8 bp for Sample Index and 58 bp for Read 2.

Mouse scRNA-seq and CITE-seq data preprocessing

The raw FASTQ files and, whenever available, cell metadata of the previously published LLC, KP, KPM, B16F10, EO771, 4T1-luc, EMT6-luc, KPC, MC38, STOSE, GL261 tumor scRNA-seq/Cite-seq datasets were downloaded from GEO, ArrayExpress, or ENA (Supplementary Data 1). For the GBM (*CAG*-LSL-*EGFR*^{WT}; InkΔ2/3$^{-/-}$;*PTEN*^{2lox}), and KPCY (*Pdx1*$^{Cre/+}$; LSL-*Kras*$^{G12D/+}$; *Trp53*$^{R172H/+}$; LSL-YFP) and the kinetic B16F10 tumor datasets, the gene expression matrices or seurat R objects were downloaded from GEO or EMBL-EBI (Supplementary Data 1).

For all the 10x Genomics newly generated datasets (EO771, PyMT, KPM, B16F10, LLC-OVA tumors; and spleen and LNs from LLC-OVA-bearing and naive mice; see Supplementary Data 1), as well as for the previously published datasets where FASTQ files were downloaded, the Cell Ranger software (10x Genomics, v.3.1.0/6.0.2/7.1.0/9.0.0) was used to perform alignment of the RNA FASTQ sequencing reads to the reference genome (Mus musculus mm10), barcode processing, and generation of the RNA and Antibody-Derived Tags (ADT) Unique Molecular Identifier (UMI) count matrices.

For the newly generated BD Rhapsody B16F10 dataset, the BD Rhapsody Sequence Analysis Pipeline (BD; version 2.0.1) was used to map the FASTQ files to the mouse reference genome (GRCm38) and generate count matrices.

The average mean number of mapped RNA reads per cell for the newly generated datasets was 29391.5 ± 5920 SE, with an average sequencing saturation metric of 53,15 ± 5.03 SE. The newly generated dataset included two CITE-seq samples (KP early and late stage), which yielded mean 4763.5 ± 1913.5 SE ADT reads per cell, and 58.8% ± 12.8% ADT sequencing saturation.

The raw RNA count matrices of all datasets included in the TADC atlas were subjected to background signal removal (ambient RNA or ADT background signal) using CellBender[166] (v0.3.0). For running CellBender, default fpr value (target false positive rate, higher fpr results in removal of more background) was used for majority of the samples (fpr = 0.01). For some samples, higher background levels were observed on the quality control barcode rank plot, which shows the library size of each sequenced droplet (as described in the CellBender usage documentation), and for those samples fpr = 0.02 was used on the RNA matrices (EO771 Clappaert et al.[59]; GL261 Pombo Antunes et al.[35]. WT2, WT3 and WT4; MC38 Hamouda et al.[65]; PyMT Jiang et al.[61]; PyMT Ye et al.[62] GSM5974493, GSM5974495 and GSM5974493; KP early stage and KP late stage; KPM; B16F10; EMT6 Carpen et al.[60]). The CellBender corrected matrices were used for further downstream data analysis.

The data from each dataset included in the TADC atlas were processed separately to detect the different MNP subpopulations in a uniform way, which would allow to assess the batch correction accuracy after integration. The MNP cells from the previously published datasets, where cell type metadata was available (GL261 Pombo Antunes et al.[35], LLC Kiss et al.[58], LLC Murgaski et al.[43], B16F10 Murgaski et al.[43], EO771 Clappaert et al.[59], MC38 Hamouda et al.[65], MC38 Zhang et al.[64], KP Maier et al.[33]), were extracted based on the available cell type annotation. For all the remaining datasets, the gene expression matrices were preprocessed and analyzed in R using Seurat v.5.1.0. The cellular barcodes, associated with low quality "empty" droplets, were filtered out using the "emptyDrops" function of the DropletUtils package v1.8.0/1.10.2/1.14.2/1.26.0 with the recommended FDR cut-off ≤0.1 for deviation from the ambient RNA profile, using the raw gene expression matrix, before correction with CellBender. Outlier cells were additionally identified based on library size, number of expressed genes and mitochondrial proportion per cell using scater v.1.22.0/1.34.0; cells were tagged as possible outliers, when they were more than three median absolute deviations distant from the median value of each metric across all cells (above the median for mitochondrial proportion and below the median for library size and number of expressed genes). A doublet score was assigned to each cell based on using scDblFinder function of the scDblFinder package v1.20.0. The gene expression matrices were further normalized, and scaled, followed by the selection of highly variable genes (HVG), principal component analysis (PCA), and unsupervised clustering using Seurat. The genes, specifically expressed in each cluster, were identified via differential expression (DE) analysis with the "FindMarkers" function of Seurat (Wilcoxon Rank Sum test). The *p*-values of DE were adjusted for multiple testing with Bonferroni correction. Clustering results were visualized with two-dimensional scatter plots using the Uniform Manifold Approximation and Projection (UMAP) dimensionality reduction method. We excluded cell populations that showed both high doublet score and expression of markers, specific for two different cell types. Additionally, we excluded clusters that were outliers due to low library size and number of expressed genes, and that lacked a distinct gene signature compared to the other clusters.

The MNP cells from each tumor dataset included in the TADC atlas were extracted and reclustered. Some of the identified clusters, for example the monocytes, were extracted and further subclustered, in order to better separate the classical, nonclassical, and differentiating immature macrophages. For the reclustering of the GL261 Pombo Antunes et al.[35] MNP dataset, batch correction was applied using harmony v.1.2.0 to ameliorate the variability between batches. The cluster annotation of the MNPs in each dataset was harmonized based on common gene expression pattern: nonclassical (*Ace, Ear2)* and classical monocytes (*Ly6c2*, *Mgst1, Chil3)*, immature macrophages *(Ccr2, C1qa*^{lo}*)*, mature macrophages (*C1qa*^{hi}*, Ms4a7, Apoe, Selenop*), hypoxic macrophages (*Vegfa, Bnip3, Hilpda*), lipid macrophages (*Gpnmb, Apoe, Fabp5, Lgals3*), Cxcl9 macrophages (*Cxcl9, Cxcl10*, Gbp3), ISG macrophages (*Isg15, Isg20, Ifit3*), Ctsk macrophages (*Ctsk, Atp6v0d2, Slc9b2, Ocstamp, Bdh2*), Retnla macrophages (*Retnla, Batf3, Cd226, Lyz1, Clec4b1*), large peritoneal macrophages (*Gata6*, *Icam2*, *Itga6*, *Timd4, Prg4, Selp, Prtn3, Cgnl1, Alox15, Padi4, Cd5l*), Cx3cr1 macrophages (*Cx3cr1, Cd72, MHCII-hi, Pmepa1, Tanc2, Vcam1*), microglia (*Tmem119, P2ry12, Sall1, Cx3cr1*), alveolar macrophages (*Car4, Ear2, Acp5, Il18, Plet1, Fabp1, Itgax, Chil3, Ctsk, Atp6v0d2, Runx2*), immature alveolar macrophages (*Itgax, Chil3, Ctsk, Atp6v0d2, Runx2, Ly6c2, Ccr2, C1qa)*, cDC2 (*Cd209a, Flt3, H2-Aa*), pre-DC (*Cd7, Sell, Apod)*, pDC (*Ccr9, Siglech, Pacsin1*), cDC1 (*Clec9a, Xcr1*), CCR7 DC (*Ccr7, Cacnb3*) and proliferating subsets (*Mki67, Stmn1*).

The DC clusters of the mouse EO771 (EO771 D9, D14 and D20) and the KP kinetic datasets (KP early and KP late stage) were subsetted and reclustered. The identity of each cluster was defined by the elaborate gene signatures identified in the mouse tumor atlas and the DE genes between the clusters. For the integrated organ dataset DC analysis, batch correction was applied using harmony v.1.2.3. The Seurat object of GBM (CAG-LSL-*EGFR*^{WT}; InkΔ2/3$^{-/-}$; *PTEN*^{2lox})[83], and the expression matrices of KPCY[82] and B16F10[84], were used for further clustering and analysis.

Reference-based mapping of Ccl5 Ccr7 DC on the LNP dataset

Reference-based mapping was performed using the Seurat R package (v5.1.0)[167] in R version 4.3.3 to inspect the DC maturation state (especially in the Ccl5 Ccr7 DC cluster) in our LLC(OVA) integrated organ scRNA-seq dataset of tumors, spleens and LNs in further detail. To accomplish this, we used a reference DC maturation CITE-seq dataset[80], containing DCs maturing under homeostatic and immunogenic conditions using lipid nanoparticles (LNP). The seurat R object of the dataset was kindly provided by the publication authors. The annotated clusters in this dataset were used for label transfer to our dataset. Anchors between the reference and our query dataset were identified using the FindTransferAnchors() function with log-normalized data, 30 principal components, and the reference PCA reduction. Cell type predictions and accompanying metadata were transferred to our query dataset using the TransferData() function. For visualization purposes, our query dataset was projected onto the

reference UMAP using the MapQuery() function. This enabled direct comparison of predicted and original annotations to determine the type of DC maturation occurring in our query dataset.

Public human scRNA-seq data preprocessing

For the melanoma tumor dataset of Brown et al.[28], and the pancreatic cancer dataset of Lin et al.[94], we extracted the raw FASTQ files from GSE137710 and GSE154778, respectively, and aligned and preprocessed the RNA data with Cell Ranger v.2.1.1/3.0.2 using the GRCh38 v.1.2.0 reference genome. For the remaining human datasets, the gene expression matrices from the human cancer datasets were downloaded from GEO or other sources as described in Supplementary Data 14. The gene names of the Zhang et al.[64] CRC dataset expression matrices and those of the Zhang et al.[100] melanoma dataset were converted to the Ensembl v.80 gene naming nomenclature (corresponding to the 10x GRCh38 v.1.2.0 reference genome) using biomartR v.2.50.3. For the datasets where cell type metadata was available, the MNP cells were extracted. For the remaining datasets, the gene expression matrices were processed as described above for the mouse tumor datasets, with the additional step of batch correction of the PCA embeddings using harmony v.1.2.0 to reduce the variability between patients.

Similar to the mouse tumor datasets, the MNP cells from each human dataset were reclustered and annotated using uniform gene signatures: nonclassical (*FCGR3A, LST1, CX3CR1)*, classical (*CD14, FCN1, VCAN)* and intermediate monocytes (gene signature intermediate between classical and nonclassical), TGFBI Mac *(TGFBI, SPP1, CLEC5A, LYZ, C1QA/B^{lo})*, mature macrophages (*C1QA/B*^{hi}, APOE, APOC1), hypoxic-macrophages (*VEGFA, BNIP3, ERO1A*), lipid macrophages (*GPNMB, CTSB, APOC1, FABP5*), FOLR2 macrophages (*FOLR2, SEPP1, SLC40A1*), HSP-macrophages (*HSPA1A, HSPA1B, DNAJB1*), ISG macrophages (*ISG20, ISG15, IFIT1, IFIT3, CXCL9*), SAA1 macrophages (*SAA1, SAA2, C3*), LYVE1 macrophages (*LYVE1, CD163*), FABP4 macrophages (*FABP4, MARCO, MCEMP1*), CTSK macrophages (*CTSK, ATP6V0D2, TNFRSF11A*), CX3CR1 Mac (*CX3CR1, C3, TREM2*), microglia (*TMEM119, CX3CR1, SALL1*), cDC2 (*FCER1A, CD1C, HLA-DPB1*), RORC DC (*RORC, ACY3, PRMD16, TMIGD2*), pDC (*LILRA4, IRF7*), cDC1 (*CLEC9A, XCR1*), CCR7 DC (*CCR7, LAMP3*), and proliferating subsets (*MKI67, STMN1*).

Data integration of the human/mouse MNP pan-cancer atlases

The generation of the human and mouse pan-cancer tumor atlases followed the pipeline, visualized in Supplementary Fig. 1. For the batch correction of the atlases, we chose to use the harmony package[67], shown to be one of the best performing methods in a recent benchmarking study[168]. The CellBender-corrected gene expression matrices of the individual MNP tumor datasets were concatenated. In order to have a more balanced distribution of cancer types in the atlas, the datasets of the most abundant cancers were randomly downsampled. In the mouse MNP atlas, the GL261 and the MC38 cells were randomly downsampled from 26,690 and 29,776 cells to the number of cells of the most abundant from the remaining tumor models (21,792 cells). In the human MNP atlas, the GBM and the lung adenocarcinoma datasets were downsampled from 54,308 and 33,847 cells, respectively, to the number of cells of the most abundant from the remaining cancer types (8782). The concatenated expression data was normalized for library size, HVG were selected, followed by data scaling for each gene and PCA dimensionality reduction. For the human tumor atlas, from the intially selected HVGs only those were retained, that were expressed in all integrated datasets, as the alignment and gene expression matrix generation was done with different software versions and different versions of the reference genome. Next, batch correction was performed on the PCA embeddings using harmony v.0.1.1/1.2.3. Harmony performs soft k-means clustering on the PCA embeddings, which assigns cells to clusters, while favoring a mixed batch representation. Then, the cluster centroids for each dataset are used to compute cluster-specific linear correction factors that correct the assignment of the cluster-weighted average of each cell. This process is iterated until all cell-cluster assignments are stable. The *theta* parameter allows controlling the degree of penalty for dependence between batch membership and cluster assignment, with higher values resulting in stronger alignment of the batches. To optimize the performance of harmony to our data, we tested 30 different parameter combinations, by varying *theta*, the number of principal components (PC) used, and three preprocessing approaches: (1) removing mitochondrial, ribosomal and immunoglobulin genes (including *HBA1, HBB, HBA2, ALB, ATP6V0C, RACK1, MALAT1, GABARAP, SFTPC* for the human atlas) from the HVG list (HVG filter, Supplementary Data 19), (2) regressing out the percentage expressed mitochondrial genes and percentage expressed ribosomal genes from each gene of the gene expression matrix before data scaling using the "ScaleData" Seurat functionality (Regression), or (3) neither of those options (Supplementary Fig. 1). We evaluated the performance of the batch correction with the different parameter combination using diagnostic UMAP plots, visualizing the batch and the cell type identity before and after correction, combined with the local inverse Simpson's index (LISI) metric using the lisi v1.0 R package[67]. LISI measures the diversity in local neighborhood of each cell. In order to assess batch alignment efficiency, we used integration LISI (iLISI), which represents the effective number of batches in a cell's neighborhood[67]. If a neighborhood is represented by only a single dataset, it will get an iLISI of 1, while a neighborhood with an equal number of cells from 2 datasets will get an iLISI of 2. For evaluating the accuracy of the batch correction, the cell type LISI (cLISI) was employed, which denotes the effective number of cell types in the neighborhood[67]. Ideally, cLISI metric should be close to 1, reflecting full separation of unique cell types, while higher values indicate presence of more than one cell type in the cell's neighborhood. The evaluation of the batch correction parameters on the MNP atlases yielded as optimal the preprocessing with HVG filter, *theta* = 1, 50 PC and *theta* = 0, 40 PC for the mouse and human tumor atlas, respectively. The corrected PC embeddings were used for downstream analysis, including clustering and UMAP. The genes or surface proteins, specific for each cluster, were identified with the "FindMarkers" function of Seurat. The *p*-values of DE genes/ surface proteins were adjusted for multiple testing with Bonferroni correction. We excluded cell populations that showed both high doublet score and expression of markers, specific for two different cell types, as well as clusters that were outliers due to low library size and number of expressed genes, and that lacked a distinct gene signature compared to the other clusters. Some of the identified clusters, including the proliferating MNPs and some DC clusters, were extracted and further subclustered to be able to better separate DCs from the monocytes and macrophage cells.

Human and mouse TADC pan-cancer atlas

For the mouse and human DC pan-cancer tumor atlas generation, all the DC clusters from the MNP tumor atlases were subsetted. HVG were selected and filtered using the same list of genes as for the MNP atlas (Supplementary Data 19). The data was scaled and used for PCA analysis, followed with batch effect correction of the PC embeddings, using *theta* = 1, 50 PC and *theta* = 0, 35 PC for the mouse and the human DC atlas, respectively. Then, the corrected PC embeddings were used for downstream analysis as described above for the MNP atlases.

After clustering the DC atlases, we observed clusters that showed macrophage and monocyte marker expression (including *C1qa/b, Adgre1, Ms4a7, Ly6c2, Chil3, Vcan* for mouse and *C1QA/B, APOE, CD14, FCN1, VCAN* for human). As the DC3 and cDC2B lineages are known to show monocyte and macrophage gene expression, we examined the doublet score, number of genes expressed per cell and library size of the identified clusters to exclude potential doublets with macrophages. The clusters with high doublet score, number of expressed

genes and library size were excluded. Additionally, we observed in the mouse DC atlas a cluster with clear bimodal distribution of the number of expressed genes per cell, indicating the presence of doublet and non-doublet populations in this cluster. We selected a threshold for the number of expressed gene per cell (3900 genes) based on its distribution in this cluster and separated the cells from this cluster with number of expressed genes above the threshold. The selected cells had also higher doublet score and library size than the remaining cells of that cluster, as well as other DC and macrophage clusters, therefore they were excluded from further analysis. Other artifact clusters with low library size, low number of expressed genes and lacking a distinct gene signature compared to the other clusters were removed as well. The remaining dataset was reclustered, following the same procedure as above, including HVG selection, PCA, batch correction, clustering, and UMAP.

Mouse LLC(OVA) tumor, spleen, and lymph node integrated dataset

All the DC clusters from the spleen and the LN datasets (LLC-OVA spleen and LN, and naive spleen and LN) were extracted and were integrated with the LLC and LLC-OVA datasets from the TADC atlas (mouse integrated organ dataset (LLC Kiss et al.[58], LLC Murgaski et al.[43], LLC-OVA tumor)). HVG were selected and filtered using the same list of genes as for the mouse tumor MNP and DC atlas (Supplementary Data 19). Next, the expression data was scaled and used for PCA analysis, followed with batch effect correction of the PC embeddings, using *theta* = 2 and 50 PC. Then, the corrected PC embeddings were used for downstream analysis as described above for the mouse tumor MNP and DC atlases. The cDC2, DC3 and progenitor subsets were subsetted and reclustered using the same procedure. The identity of each cluster was defined by the elaborate gene signatures identified in the mouse tumor atlas, the DE genes between the clusters, and splenic DC gene signatures from the Liu et al.[2] and Rodrigues et al.[3] naive splenic DC datasets (pDC-like: *Siglech, Irf8, Cd8a, Gria3, Selenop*; cDC2: *Ltb, H2-DMb2, H2-Oa, Ffar2, Ppp1r14a, Rgs2, Mdh2; Rorc DC: Rorc, Nrgn, Sox4, Ltb, Tcf7, Prnp, Nedd4, Tmod2*; cDC2B: *Itgax, Csf1r, Clec4a3, Sirpb1a/c, S100a11, H2-DMb2, H2-Oa, Ffar2,Ltb*; Pro-DC3: *Ccl6, Ccl9, Cd209a, Itgb7, Anxa1, Cd300lg*; DC3: *Csf1r, Clec4a3, Lyz2, Tgfbi, Adgre4, Ccl6, Ccl9*).

Clustering significance testing

The clustering results of the mouse and human TADC atlas were subjected to a significance test using the sc-SHC[78] package v. 0.1.0. The sc-SHC algorithm hierarchically clusters the centers of clusters, provided by the user and performs recursive statistical testing at each node of the resulting hierarchical tree. The *p*-values are corrected for multiple, sequential hypothesis testing and controlled for the family-wise error rate. In order to account for the batch effect between datasets, the harmony batch-corrected PCA embeddings for each dataset were extracted and scaled to integer values by min-max normalization for each cell between 0 and the 99 quantiles of the raw counts of the respective dataset. The clustering significance was subsequently tested on the modified PCA embeddings.

RNA Velocity analysis

The spliced and unspliced counts per gene for each TADC atlas dataset were estimated using velocyto[76] v.0.17.17 from the raw BAM files generated from Cell Ranger or BD Rhapsody Sequence Analysis Pipeline. For the 10x Genomics datasets, the run10x velocyto command was used, while for the BD Rhapsody dataset a publicly available Nextflow workflow (https://github.com/grst/bd_rhapsody_velocyto) was first used to make the BAM file compatible with velocyto https://github.com/grst/bd_rhapsody_velocyto, followed by using the velocyto run command. Next, RNA velocity was estimated with the velocyto.R_0.6 package using gene-relative model with $k = 20$ cell kNN pooling and using top/bottom 2% quantiles for gamma fit ("fit.quantile" option). The RNA velocities were visualized on the UMAP embeddings via correlation-based transition probability matrix within the kNN graph using neighborhood size of 50 or 20 cells, and grid.$n = 70$ or 25 for the full mouse TADC atlas and the Pre-cDC1/cDC1/Ccr7 Cxcl9 cDC1/ Ccr7 cDC1/Ccr7 DC subset of the TADC atlas, respectively.

Gene signature scores

The gene signature scores used for annotation of the TADC atlases were calculated based on differentially expressed genes between key DC populations from public scRNA-seq and bulk RNA-seq datasets (Supplementary Data 6, 14). The expression data from each dataset was downloaded as described in Supplementary Data 6, 14. The genes for the signature scores based on scRNA-seq datasets were identified with the "FindMarkers" function of Seurat. For the noncanonical DC score based on the bulk RNAseq dataset of Lau et al.[21], the raw expression counts were downloaded from GSE76132, and the genes for the signature were selected by performing differential expression analysis with DESeq2 v1.30.1. Thresholds for *p*-value and log fold change were applied for the genes included for each signature, as shown in Supplementary Data 6, 14. The genes for the Minutti et al.[29] bulk RNAseq pre-cDC2A versus pre-cDC2B score were taken from the Supplementary Table 3 of Minutti et al. For the human blood DC and monocyte scRNA-seq dataset of Dutertre et al.[30] there was no cell type labels available, therefore we preprocessed and clustered the gene expression matrix using the same approach as described for the mouse scRNAseq and CITE-seq data preprocessing. The clusters were annotated based on cell type markers described by Dutertre et al.[30].

The signature scores were calculated for each cell in the respective datasets, using the "AddModuleScore" function of Seurat. This function yields the average expression level for a group of genes of interest, subtracted by the average expression of a control gene set. All analyzed genes are binned based on average expression, and the control genes are randomly selected from each bin.

The interferon and maturation gene signatures in the human TADC subsets, related to Fig. 7C, D, were based on the average expression of (1) selected Interferon-stimulated and inflammation-associated genes (*IFIT3, IFIT2, IFIT1, ISG20, ISG15, RSAD2, IRF7, STAT1, MX1, CCL4, OAS1A, IFITM1, IFITM3, CXCL10, CXCL9*); (2) selected DC-activation and maturation genes (*CD80, CD83, CD40, CD86, FSCN1, BIRC3, LAMP3, CCL19, LAD1, MARCKS, TNFAIP2, CCR7, CCL22, MARCKSL1, EBI3, TNFRSF11B, NUB1, INSM1, RAB9A, LY75, SIAH2, POGLUT1, KDM2B, MGLL*[4,39]).

Single-cell regulatory network inference with SCENIC

We performed single-cell regulatory network inference analysis using SCENIC[169] v1.3.1, RcisTarget_v.1.28.1 and AUCell_v.1.30.1, using as input the raw, untransformed RNA UMI counts of the mouse and the human tumor DC atlases. The co-expression network was generated using GRNBoost2 via arboreto v0.1.6. For running GRNBoost2, the expression matrices were filtered for genes with over 30 or 20 UMI counts and expressed in at least 20 or 10 cells in the mouse and human dataset, respectively. The resulting transcription factor by gene targets matrices were imported in R and further analyzed with the SCENIC workflow with default parameters. The regulon activity, which identifies and scores gene regulatory networks or regulons in single cells, was calculated using AUCell as previously described[169]. The better the gene targets of a regulon match the highly expressed genes of a certain cell, the higher the AUC value (also named regulon activity) of the regulon in that particular cell.

Gene expression and regulon activity correlation plots

For the gene expression correlation plot in Supplementary Fig. 9F from the LLC(OVA) integrated organ dataset, 2000 shared variable genes were selected from the respective datasets using the

"SelectIntegrationFeatures" function of Seurat. This function ranks genes by the number of datasets they are deemed variable in and returns the top scoring genes by this ranking. The expression values for the selected genes were scaled per gene using the "ScaleData" Seurat function. Next average expression was calculated for the shared variable genes in selected clusters in the two datasets being compared using the "AverageExpression" function of Seurat. Finally, Pearson correlation coefficient was computed for each corresponding cluster in the two datasets using the average expression values for the selected genes.

For the gene expression correlation plots between mouse and human DC clusters in Fig. 6A and Supplementary Fig. 12A, the gene symbols of the genes expressed in the human and mouse DC atlas were converted between human and mouse based on one-to-one orthology using biomartR v2.64.0. Then, 2000 shared HVG were selected using Seurat's "SelectIntegrationFeatures" function, and their scaled expression was averaged per cluster and Pearson correlation coefficients were calculated between the corresponding mouse and human DC clusters. For Fig. 6A, B, all human cDC2B clusters, human DC3 clusters, the mouse cDC2A, mouse ISG-cDC2, and mouse ISG-DC3 clusters were merged prior to averaging. Additionally, the human CXCL10 cDC2 cluster was merged with human ISG-cDC2, the mouse Pre-cDC1 – with the cDC1 cluster, the mouse ISG-Pro-DC3 with the ISG-DC3, mouse Ccr7 ISG-cDC2 with the ISG-cDC2 and the mouse Ccr7 Pre-cDC2A and Ccr7 cDC2A – with Ccr7 cDC2.

For the mouse-human regulon activity correlation plots in Fig. 6B and Supplementary Fig. 12B, the name of each human regulon was converted to the corresponding mouse transcription factor based on one-to-one orthology using biomartR. Next, 100 highly variable regulons were identified using the "SelectIntegrationFeatures" function, the regulon activity was scaled per regulon, and the average regulon activity per cluster was calculated for each regulon. Finally, Pearson correlation coefficients were calculated between the corresponding mouse and human DC clusters, using the average scaled regulon activity.

Differential cell abundance analysis between tumor types with MiloR

For assessing the differential abundance of the distinct cell clusters in the human and mouse TADC atlases between different tumor types/models, we utilized MiloR[170] v.2.4.1. MiloR is a scalable statistical framework, which tests for differential abundance by assigning cells to partially overlapping neighborhoods on a *k*-nearest neighbor (KNN) graph. MiloR uses a negative binomial generalized linear model and computes two-sided *p*-values to determine if cell abundance increases or decreases between conditions. The *p*-values are corrected for multiple hypothesis testing via the "spatial FDR" approach, which adjusts *p*-values based on the amount of overlap between neighborhoods. The KNN graph was constructed using 20 nearest neighbors for the harmony corrected PCA embeddings with the buildGraph() function. Representative neighborhoods were subsequently defined with the makeNhoods() function using 20 nearest neighbors and 50 harmony corrected PCA dimensions. Finally, differential abundance tests were completed between each tumor type and all the remaining tumor types in the dataset. Only tumor models/types containing more than two mouse samples/human patients were included in the analysis. Each cell neighborhood was annotated to the cell cluster present with the highest frequency in this neighborhood. Neighborhoods, where less than 70% of the cells are part of the most frequent cell cluster, were annotated as "Mixed".

Survival analysis

To investigate the impact of our human TADC subsets/states on cancer patient survival, we applied SCIPAC[103], an algorithm that infers a quantitative estimation of cell-phenotype association by matching a single-cell dataset and a bulk dataset with the phenotype of interest. We performed this analysis on the 9,618 DCs to obtain the association of cDC1, pDC-like, and proliferating cDC1 cells and the 6,719 sub-clustered cDC2/DC3 in combination with the progression-free survival of 13 large TCGA datasets: Bladder Urothelial Carcinoma (blca), Breast Invasive Carcinoma (brca), Colorectal Adenocarcinoma and rectal adenocarcinoma (coadread), Head and Neck Squamous Cell Carcinoma (hnsc), Brain Lower Grade Glioma (lgg), Lung Adenocarcinoma (luad), Lung Squamous Cell Carcinoma (lusc), Stomach Adenocarcinoma (stad), Thyroid Carcinoma (thca), Uterine Corpus Endometrial Carcinoma (ucec), Liver Hepatocellular Carcinoma (lihc), High-Grade Serous Ovarian Cancer (hgsoc), Skin Cutaneous Melanoma (skcm). The TCGA datasets and their clinical data were downloaded using cBioportal[171]. In order to correct for the effect of confounding factors in the TCGA datasets we slightly modified the SCIPAC code in a way we could include covariates in the Cox regression step. The covariates used in the TCGA datasets (whenever enough samples were available) were: age, (FIGO) stage, grade, remaining tumor after surgery, adjuvant (radiation) therapy, and subtype. To obtain an overall association for each subtype, a meta-analysis was performed by the R package metafor[172].

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The generated human and mouse atlases described in this article can be accessed via our interactive webserver www.single-cell.be/Laouimmunology/TumorDCAtlas. This tool facilitates the evaluation of gene and surface protein (wherever available) expression across the different datasets generated in this manuscript. Additionally, the newly generated mouse scRNA-seq/Cite-seq raw data and gene expression matrices generated in this study are deposited at GEO (NCBI) under accession number GSE304727. The data sources of all the previously published datasets used in this study are listed in Supplementary Data 1 and 14. The remaining data supporting the findings of this study are available within the Article, the Supplementary Information, or the Source data file. Source data are provided with this paper.

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Acknowledgements

We thank Nadia Abou, Eleonora Omasta, Ellen Vaneetvelde, and Maité Schuurmans from Vrije Universiteit Brussel for administrative and technical assistance. We would like to thank the VIB Nucleomics Core and the VIB Single Cell Core for providing access to RNA sequencing technologies, their support, and access to the instrument park (vib.be/core-facilities). The computational resources and services used for the patient overall survival analysis in this work were provided by the VSC (Flemish Supercomputer Center), funded by the Research Foundation—Flanders (FWO) and the Flemish Government—department EWI. The results of the SCIPAC survival analysis are in whole or part based upon data generated by the TCGA Research Network: https://www.cancer.gov/tcga. Schematic figures were generated using BioRender.

Author contributions

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Funding

A.A.C. is supported by a predoctoral grant from FWO Vlaanderen (1169521N) and an Emmanuel Van der Schueren Grant of Kom op tegen Kanker. P.M.R.B. is supported by a predoctoral grant from FWO Vlaanderen (1154720N). E.H. is supported by a postdoctoral grant from FWO Vlaanderen (12Y1922N). S.D. is supported by a postdoctoral grant from Stichting tegen kanker (2021-023). A.B. and P.C. are FNRS Senior Research Associates. M.D.P. was supported by a grant from a generous donor advised by Carigest SA. A.C. is supported by Kom Op Tegen Kanker, FWO, KU Leuven, Vriendtjes tegen Kanker, HEBE foundation, Leuven Cancer Institute, and Olivia fund. Da.L is supported by grants from FWO (G053321N), Kom op tegen Kanker, Stichting tegen kanker, VIB and Vrije Universiteit Brussel.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41467-026-73721-w.

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Peer review information *Nature Communications* thanks Marc Dalod and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

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